

AutoMate: A Dataset and Learning Approach for Automatic Mating of CAD Assemblies

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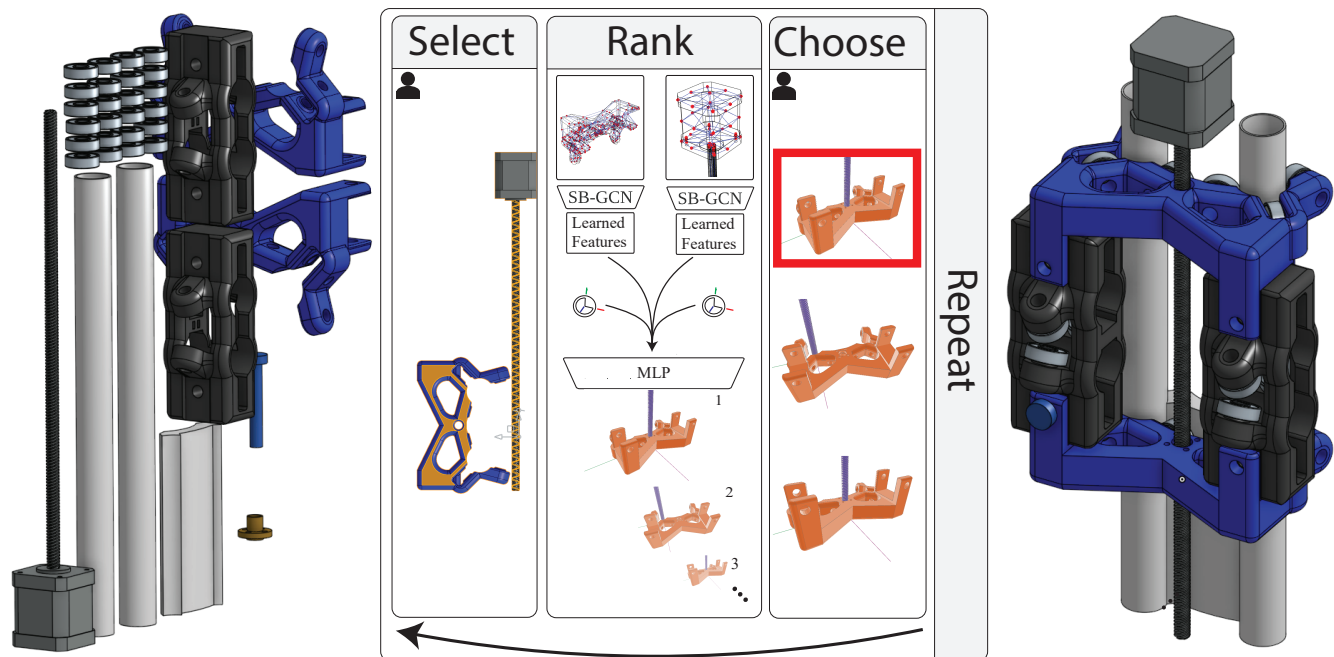


Fig. 1. Our tool assists a designer in assembling a collection of CAD parts (left) into a functional assembly (right). First user selects a pair of parts to mate, and indicates roughly where the parts should be mated by where they click on each part. SB-GCN then creates embeddings for each topological entity of the selected parts' BREPs, which are used as input to a classifier that scores potential mates defined in reference to these topological entities. Our system presents a list of suggestions that the user can pick from. This cycle is repeated until a working mechanical assembly is realized.

Assembly modeling is a core task of computer aided design (CAD), comprising around one third of the work in a CAD workflow. Optimizing this process therefore represents a huge opportunity in the design of a CAD system, but current research of assembly based modeling is not directly

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applicable to modern CAD systems because it eschews the dominant data structure of modern CAD: parametric boundary representations (BREPs). CAD assembly modeling defines assemblies as a system of pairwise constraints, called *mates*, between parts, which are defined relative to BREP topology rather than in world coordinates common to existing work. We propose SB-GCN, a representation learning scheme on BREPs that retains the topological structure of parts, and use these learned representations to predict CAD type mates. To train our system, we compiled the first large scale dataset of BREP CAD assemblies, which we are releasing along with benchmark mate prediction tasks. Finally, we demonstrate the compatibility of our model with an existing commercial CAD system by building a tool that assists users in mate creation by suggesting mate completions, with 72.2% accuracy.

CCS Concepts: • **Computing methodologies** → **Parametric curve and surface models; Machine learning.**

AutoMate：用于CAD装配件自动配对的数据集与学习方法

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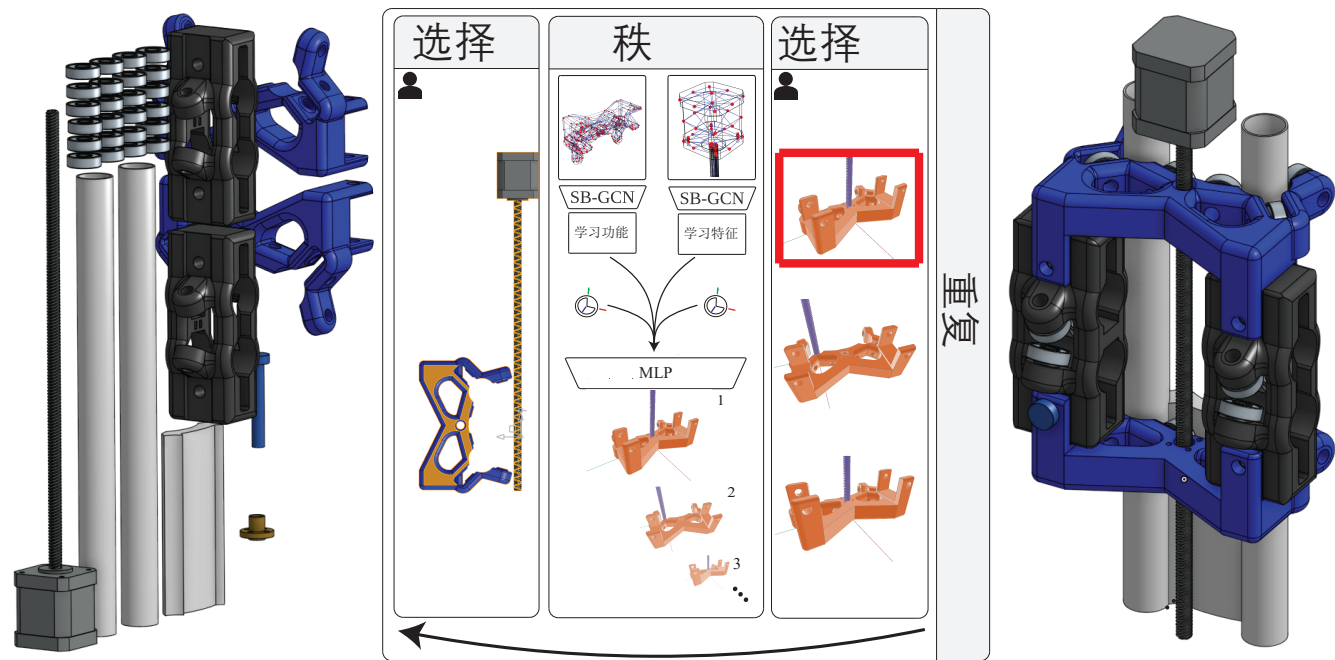


图1. 我们的工具可协助设计师将一组CAD零件（左图）组装成一个功能性装配体（右图）。首先，用户需选择一对要配合的零件，并通过在各零件上点击来大致标示这些零件应配合的位置。随后，SB-GCN 会为所选零件BREP模型中的每个拓扑实体生成嵌入向量，这些嵌入向量将作为输入提供给分类器，用于对基于这些拓扑实体定义的潜在配合方案进行评分。系统会呈现一份建议列表供用户选择。此循环过程将持续重复，直至完成一个可用的机械装配体。

装配建模是计算机辅助设计（CAD）的核心任务，约占CAD工作流程中总工作量的三分之一。因此，优化这一过程为CAD系统的设计提供了巨大的机遇，但目前关于基于装配的建模研究尚未直接针对此目标展开。

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该方法适用于现代CAD系统，因为它摒弃了现代CAD领域主流的数据结构——参数化边界表示（BREP）。CAD装配建模将装配体定义为零件之间的一对约束关系体系（称为配合件），这些配合关系是相对于BREP 拓扑结构而非现有工作中常用的世界坐标系来定义的。我们提出了SB-GCN 这一基于BREP的表示学习方案，该方案能够保留零件的拓扑结构，并利用这些学习得到的表示来预测CAD类型的配合件。为训练我们的系统，我们编制了首个大规模BREP CAD装配数据集，并将其与基准测试配合件预测任务一并发布。最后，我们通过开发一款能通过建议配合件补全方案来辅助用户创建配合件的工具，验证了我们的模型与现有商业CAD系统的兼容性，准确率达到72.2%。

中国化学会概念：计算方法论→参数曲线与曲面模型；机器学习。

Additional Key Words and Phrases: computer-aided design, boundary representation, representation learning, assembly-based modeling

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1 INTRODUCTION

Manufactured objects are designed in parametric computer-aided design (CAD) systems as individual mechanical parts that are assembled into functional objects. In CAD, creating an assembly from parts is a time-consuming manual process of *mating* the parts, which requires both careful positioning of parts in reference to one another, as well as specification of how these parts can move relative to one another. In practice, users spend roughly one third of their time in modern CAD tools in assembly work [Onshape 2020]. In this work we develop novel machine learning techniques for predicting how to mate parts. We further integrate these predictive models into an existing commercial CAD product [PTC Inc. [n.d.]], making it easier to create mates.

While many prior techniques use machine learning in assembly-based 3D modeling tools [Huang et al. 2020; Lin et al. 2018; Mo et al. 2019a, 2020a; Sung et al. 2017; Wu et al. 2019; Yin et al. 2020; Zhu et al. 2018], these techniques are not suited for a CAD workflow. Established commercial CAD systems are built around boundary representations (BREPs, Figure 2), a data structure that represents the structure of a shape as *topological entities* of different spatial dimensions (2D faces, 1D edges and loops of edges, and 0D vertices), and explicitly captures the *analytic* geometry of each as a parametric equation, as well as the relations between topological entities. BREP based CAD systems model assemblies as a collection of pairwise constraints between parts called *mates*, where the constraints are defined relative to the topological entities. Using an analytic shape representation is crucial for the high modeling accuracy required for fabrication and integration with computed aided manufacturing (CAM). Modeling as a system of constraints enables capturing complex part interactions of mechanical motion, and allows the model to adapt globally to local changes. Defining these constraints from the topological entities allows mates to have analytic precision, as well as adapt to some parametric changes in geometry. In contrast to these CAD native assemblies, many existing assembly modeling tools use segmented meshes or point clouds to represent geometry, and predict world-coordinate, often static positions to construct assemblies, making them ill-suited for integration into the CAD design process.

While some work uses pairwise constraints [Huang et al. 2020; Lin et al. 2018; Mo et al. 2019a], to the best of our knowledge no existing assembly modeling systems create topologically defined mates, nor do any operate on BREP CAD data. Two key challenges impede the development of such a system; the lack of BREP learning models suitable for mating, and a lack of data for training such a system. Concurrent work on learned models for BREPs [Lambourne et al. 2021; Willis et al. 2020] operate on a simplified *homogeneous* BREP structure and only learn representations for faces. BREPs can be mated relative to any class of topological entity, so the mating

problem requires a *heterogeneous* BREP representation. Compiled BREP datasets [Koch et al. 2019; Lambourne et al. 2021; Willis et al. 2020] are focused on BREP modeling and do not contain an mate data, and assembly modeling datasets [Mo et al. 2019b] do not have BREP data or mates, so no suitable dataset exists for training such a model.

To overcome these challenges, we offer methodological and data contributions. First, we propose *Structured BREP Graph Convolution Networks* (SB-GCN), a structured graph representation and message passing network for BREPs that is the first heterogeneous learned BREP representation in that it learns embeddings for multiple classes of topological entity. Second, we scraped a large collection of raw BREP assemblies, and cleaned it by de-duplicating mates and assemblies and converting to a canonical format, to create the first BREP assembly and mating dataset, which we are making publicly available. We apply SB-GCN to BREP mating by modeling mates as translational and rotational constraints between *mating coordinate frames* (MCFs) – coordinate frames on BREPs that inherit their origin and alignment by reference to separate topological entities on each BREP (see Figure 3).

We demonstrate that our approach is directly applicable to existing commercial CAD system by implementing a mate recommendation system as an extension to the Onshape CAD system. Our mate model maps directly onto 180,102 of the mates in our dataset, which we use to train a ranker for potential mates and a classifier for mate degrees of freedom. Our model suggests a verifiably correct mate location 72.2% of the time on this data, significantly outperforming traditional discrete geometry approaches (46.3%).

2 RELATED WORK

Assembly-based Modeling. Assembly-based, or compositional, modeling tools create 3d models by aggregating parts from 3d shape collections. Modeling by example [Funkhouser et al. 2004] opened this field with two main subproblems; how to choose which parts to combine and how to combine them. The first problem is commonly solved by text search [Funkhouser et al. 2004], similarity search [Chaudhuri and Koltun 2010; Funkhouser et al. 2004], or consistent segmentation [Chaudhuri et al. 2011]. In our work, we assume that retrieval is solved and focus solely on the second problem of mating parts.

Traditionally, aligning parts has been performed analytically; using iterative closest points [Funkhouser et al. 2004; Sharf et al. 2006], surface registration [Chaudhuri and Koltun 2010], or energy optimization [Gonzalez and van Kaick 2018]. Newer data-driven methods learn to align shapes. In contrast to our work, most approaches [Huang et al. 2020; Jones et al. 2020; Mo et al. 2020b; Sung et al. 2017; Wu et al. 2020b; Yang et al. 2020; Yin et al. 2020] predict or generate global offsets for each part rather than pairwise constraints. Assemblies defined by global positions cannot handle replacement of parts, parametric variation of parts, or modeling degrees of freedom. Those that do model assemblies by pairwise positioning [Lin et al. 2018; Mo et al. 2019a, 2020a; Wu et al. 2019; Zhu et al. 2018] predict or generate explicit offsets and rotations, forfeiting the precision and adaptability to parametric changes that our system acquires by defining these relative to topological features.

12月。其他关键词及短语：计算机辅助设计、边界表示、表示学习、基于装配的建模

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1 引言

制造对象是在参数化计算机辅助设计（CAD）系统中作为独立机械部件进行设计的，这些部件最终会被组装成具有功能性的整体产品。在CAD领域，从零部件构建装配体是一个耗时的手动流程——即需要对零部件进行**配合**，这不仅要求精确确定各部件之间的相对位置关系，还需明确它们之间的相对运动方式。实际上，用户在现代CAD工具中的时间约有三分之一用于装配操作[Onshape 2020]。本研究开发了新型机器学习技术以预测零部件的配合方式，并将这些预测模型集成到现有的商用CAD软件中[PTC公司]。[n.d.]，从而更轻松地创建配对对象。

尽管许多先前的技术在基于装配的三维建模工具中运用了机器学习[Huang等人2020；Lin等人2018；Mo等人2019a、2020a；Sung等人2017；Wu等人2019；Yin等人2020；Zhu等人2018]，但这些技术并不适用于CAD工作流程。成熟的商业CAD系统均围绕边界表示（BREPs，图2）构建——这是一种将形状结构表示为不同空间维度的**拓扑实体**（如二维面、一维边与边环、以及零维顶点）的数据结构，并能通过参数方程明确捕捉每个实体的**解析几何特性**及其拓扑实体间的关联关系。基于BREPs的CAD系统则将装配体建模为零件间成对约束的集合，这些约束被称为**配对约束**，其中约束条件均相对于上述拓扑实体定义。采用解析形状表示对于满足制造及与计算机辅助制造（CAM）集成所需的高建模精度至关重要。将建模视为约束系统，能够捕捉机械运动中复杂的零件相互作用，并使模型能够全局适应局部变化。通过从拓扑实体定义这些约束，可使配合部件具备解析精度，并能适应几何结构中的某些参数化变化。与这些CAD原生装配方案不同，许多现有装配建模工具采用分段网格或点云来表示几何结构，并预测世界坐标系下的位置（通常为静态位置）以构建装配体，这使得它们难以适配于CAD设计流程的集成。

尽管已有部分研究采用成对约束[Huang等人2020；Lin等人2018；Mo等人2019a]，但据我们所知，目前尚无任何现有的装配建模系统能够创建拓扑定义的配对对象，也无任何系统能处理BREPs CAD数据。阻碍此类系统开发的两大关键挑战在于：缺乏适用于配对的BREPs学习模型，以及缺乏用于训练此类系统的数据。针对BREPs的已开展的研究[Lambourne等人2021；Willis等人2020]均基于简化的**异质BREPs**结构运行，且仅能学习面的表示。BREPs可相对于任意类别的拓扑实体进行配对，因

此配对问题需要一种**异质BREPs**表示方式。已编译的BREPs数据集[Koch等人2019；Lambourne等人2021；Willis等人2020]主要聚焦于BREPs建模，且未包含配对数据；而装配建模数据集[Mo等人2019b]并未获得BREPs数据或配对样本，因此不存在适用于训练此类模型的合适数据集。

为克服这些挑战，我们提供了方法论和数据层面的贡献。首先，我们提出了**结构化BREPs图卷积网络（SB-GCN）**——这是一种针对BREPs的结构化图表示及消息传递网络，作为首个异构学习型BREPs表征方法，能够为多种拓扑实体类别学习嵌入向量。其次，我们收集了大量原始BREPs组装数据，并通过去重处理配对序列与组装体、转换为标准格式等方式进行清洗，构建了首个BREPs组装与配对数据集，并将其公开发布。我们将SB-GCN应用于BREPs配对分析时，将配对序列建模为**配对坐标系（MCFs）**之间的平移与旋转约束——这些坐标系基于BREPs上的独立拓扑实体确定其原点并实现对齐（见图3）。

我们通过在Onshape CAD系统中集成部件推荐系统作为扩展模块，证明了该方法可直接应用于现有商用CAD系统。我们的部件模型可与数据集中的180,102个部件直接对应，利用这些数据训练了潜在部件的排序模型及部件自由度分类器。在该数据集上，我们的模型能够以72.2%的准确率预测出可验证正确的部件位置，显著优于传统离散几何方法（46.3%）。

2 相关工作

基于装配的建模。基于装配或组合式建模工具通过聚合来自三维形状集合的部件来创建三维模型。基于示例的建模[Funkhouser等人2004]开创了这一领域，提出了两个主要子问题：如何选择要组合的部件以及如何将它们进行组合。第一个问题通常通过文本搜索[Funkhouser等人2004]、相似性搜索[Chaudhuri与Koltun 2010；Funkhouser等人2004]或一致分割[Chaudhuri等人2011]来解决。在我们的工作中，我们假设检索问题已得到解决，仅专注于第二个问题——部件的匹配。

传统上，部件对齐通常通过解析方法实现：使用迭代最近点[Funkhouser等人2004；Sharf等人2006]、表面配准[Chaudhuri与Koltun 2010]或能量优化[Gonzalez与van Kaick 2018]。较新的数据驱动方法则致力于学习形状对齐。与我们的工作不同，大多数方法[Huang等人2020；Jones等人2020；Mo等人2020b；Sung等人2017；Wu等人2020b；Yang等人2020；Yin等人2020]仅预测或生成每个部件的全局偏移量而非成对约束条件。基于全局位置定义的装配体无法处理部件更换、部件参数化变化或建模自由度问题。而采用成对定位法建模装配体的方法[Lin等人2018；Mo等人2019a、2020a；Wu等人2019；Zhu等人2018]通过预测或生成显式的偏移量和旋转参数，牺牲了我们的系统通过将这些参数相对于拓扑特征定义而获得的精确度与适应性。

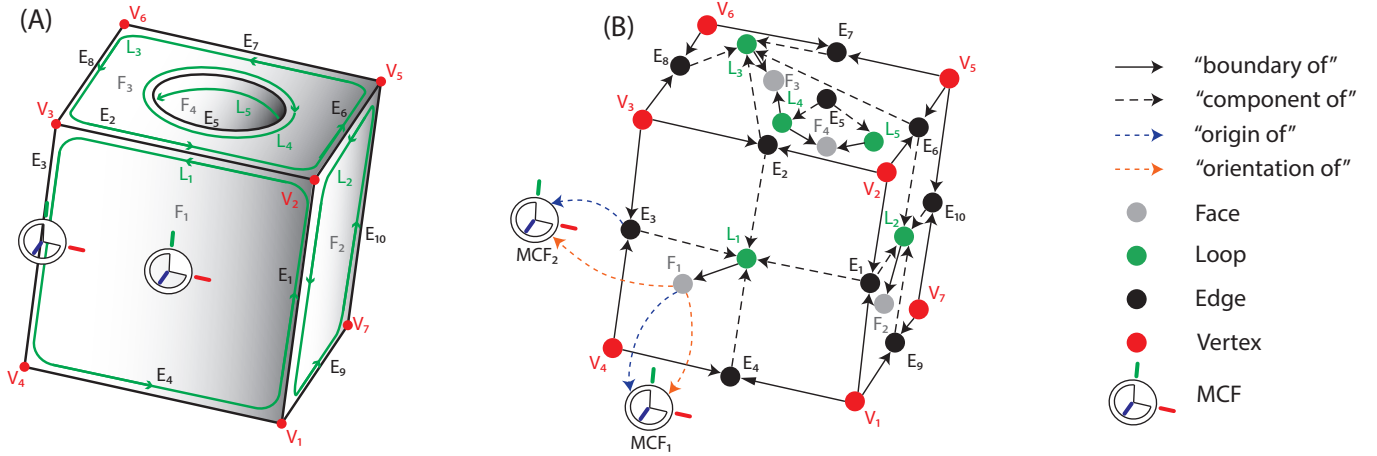


Fig. 2. Our data model. Left: A simple part modeled as a BREP, with the topological entities (faces, loops, edges, and vertices) labeled. Also shown are two mating coordinate frames, located at the centroid of F_1 and the center of E_3 . Right: Our graph representation of a BREP; a subset of the graph for the part on the left, limited to visible entities, is shown. The topological entities (graph nodes) are connected via boundary or component relations (graph edges). This graph has consistent structure; graph edges only go vertex-to-edge, edge-to-loop, or loop-to-face. Mating Coordinate Frames are defined by reference to topological entities for their location and orientation.

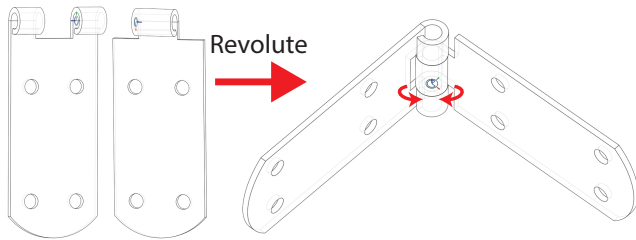


Fig. 3. In a mate, the references to topological entities specify the location constraints, and the mate type (revolute) determines the degrees of freedom.

Another limitation of these methods is that they work with non-parametric surface representations (meshes, pointclouds, voxel grids) or bounding boxes, and encode the geometry using a geometric learning model such as PointNet [Qi et al. 2017a], PointNet++ [Qi et al. 2017b], PCPNet [Guerrero et al. 2018], or DGCNN [Wang et al. 2019]; we refer the reader to Bronstein et. al [2017] for a survey geometric learning methods. Unlike these techniques, our method uses BREPs as input which provides additional CAD context and enables it to interface with CAD systems.

Our work overcomes limitations of existing assembly modeling methods in the CAD context. Existing methods are trained over categories of shapes, limiting their usefulness for general design problems. Since we learn local representations on the underlying part structure, we can avoid dependence on global part context. Additionally, none of these systems model the degrees of freedom of connections. Fab-by-example [Schulz et al. 2014] is the closest existing work to our system in capabilities; its pairwise connections are parametrically defined to adapt to parametric variation, it models the degrees of freedom of connections, and is not limited to a single

category. However, it relies on a database of hand-annotated parts and so does not generalize to arbitrary CAD data.

Learning with CAD Data. Recently, interest in data-driven approaches on parametric CAD data has increased with the creation of several CAD datasets. Koch et. al released the ABC dataset [Koch et al. 2019], a collection of CAD models sourced from Onshape, to spur development of learning on CAD data. Sketchgraphs [Seff et al. 2020] is a dataset that collects 2D sketch information from Onshape. That work also presents an application in learning the constraint graph underlying these sketches. Recently, [Willis et al. 2020] compiled a dataset of sketch-extrude modeled parts for learning 3D construction sequences, and [Cao et al. 2020] proposed a method of generating CAD datasets with segmentation annotations, and a message passing network for segmentation. Concurrent to our work, [Lambourne et al. 2021] and [Jayaraman et al. 2021] propose message passing networks to learn BREP face embeddings. Similar to this paper, these works use message passing graph neural networks to model the CAD data, but they only learn to embed one type of topology; sketches in the 2d case and faces in the 3d works. Our mating task requires us to have representations for faces, loops, edges, and vertices, since all 4 classes of topological entity can be used to mate parts. This means our graph representations are inherently heterogeneous, in contrast to the homogeneous graph representations used in prior works. These datasets and applications are centered around the part modeling part of the CAD workflow. An assembly database, PartNet [Mo et al. 2019b], does exist, and was used to train several of the assembly modeling systems mentioned earlier, but it is based around mesh geometry, not CAD models. Ours is the first dataset and application that captures assembly information - the relationships between parts.

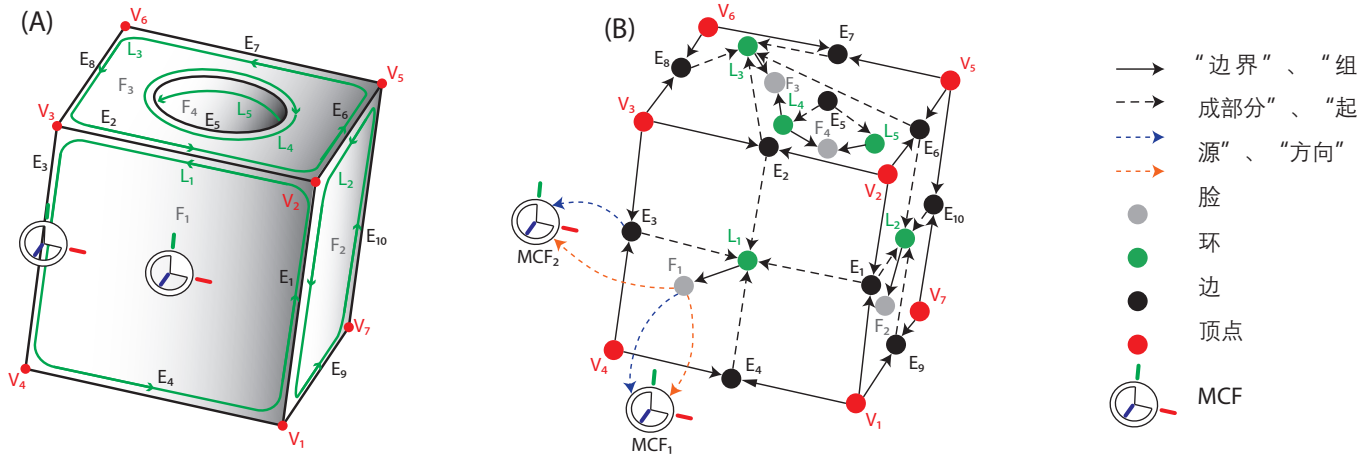


图2. 我们的数据模型。左图：一个简单的零件，以 BREP 形式建模，其拓扑实体（面、环、边和顶点）均标有标签；同时显示了两个配对坐标系，分别位于 F_1 的质心处和 E_3 的中心处。右图：我们对 BREP 的图表示；展示了左侧零件对应的图子集，该子集仅包含可见实体；其中的拓扑实体（图节点）通过边界或组件关系（图边）相互连接。此图具有一致的结构：图边仅可从顶点指向边、从边指向环或从环指向面。配对坐标系是参照以下内容定义的。用于表示其位置与朝向的拓扑实体。

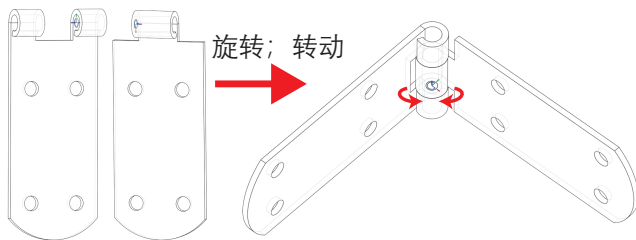


图3：在配对关系中，对拓扑实体的引用用于指定位置约束，而配对类型（旋转配对）则决定了自由度。

这些方法的另一局限性在于，它们仅能处理非参数曲面表示（网格、点云、体素网格）或边界框，并利用几何学习模型（如 PointNet[Qi 等 2017a]、PointNet++[Qi 等 2017b]、PCPNet [Guerrero等2018]或 DGCNN [Wang等2019]）来编码几何结构；关于几何学习方法的综述，可参阅Bronstein等[2017]。与这些技术不同，我们的方法采用BREP作为输入，这不仅提供了额外的CAD上下文信息，还使其能够与CAD系统进行交互。

我们的研究克服了现有CAD环境中装配建模方法的局限性。现有方法通常基于特定形状类别进行训练，这限制了其在通用设计问题中的适用性。而我们通过学习底层零件结构的局部特征表示，能够避免依赖全局零件上下文信息。此外，这些系统均未对连接部位的自由度进行建模。以Fab-by-example[Schulz 等人2014]为例，其功能最接近我们的系统：该方法采用参数化方式定义成对连接关系以适应参数变化，能准确建模连接部位

的自由度且不受单一类别限制；但因其依赖人工标注零件数据库，无法推广至任意CAD数据场景。

基于CAD数据的学习。近年来，随着多个CAD数据集的创建，对参数化CAD数据上数据驱动方法的关注度日益增加。Koch等人发布了ABC数据集[Koch等2019]——该数据集收录了源自Onshape的CAD模型，旨在推动CAD数据上的学习研究。Sketchgraphs[Sef等2020]是一个从Onshape收集二维草图信息的数据集，该研究还展示了在这些草图基础上学习其底层约束图的应用场景。近期，[威利斯 埃特；埃特 al. 2020] 研究人员编译了一套基于草图-拉伸建模的零件数据集，用于学习三维装配序列；[Cao等人2020]提出了一种生成带有分割标注的CAD数据集的方法，并设计了一个用于分割的任务的消息传递网络。与我们的工作同步，[Lambourne等人2021]和[Jayaraman等人2021]提出了消息传递网络来学习 BREP 面嵌入。与本文类似，这些研究均采用消息传递图神经网络来建模CAD数据，但它们仅能学习嵌入一种类型的拓扑结构——即二维情况下的草图以及三维情况下的面。我们的装配任务要求我们能够获得面、环、边和顶点的表征，因为这四种拓扑实体均可用于零件装配。这意味着我们的图表示本质上是异构的，与先前工作中使用的同质图表示形成对比。这些数据集及应用均围绕CAD流程中的零件建模环节展开。确实存在一个装配数据库PartNet[Mo 等人2019b]，该数据库已被用于训练多项装配模型。上述提到的MBLY建模系统均基于网格几何结构而非CAD模型。我们的数据集及应用方案是首个能够捕捉装配信息（即部件间关系）的解决方案。

Graph Network Representation Learning. Generic graph-based learning techniques have been applied to geometry, part assemblies, social networks, and other types of data. In this work we leverage these techniques to process CAD data by applying them to a graph structure that takes advantage of the structure of BREPs. We refer the reader to two recent surveys for a more in-depth overview of the field, [Wu et al. 2020a], [Zhang et al. 2020] and only reference the most relevant works here. Most modern works follow the Graph Convolutional Approach (GCN) of Kipf and Welling [2017], which generalizes the Euclidean convolution to graph domains via a two step message passing process: features of neighboring nodes are collected as computed messages, then these messages are aggregated locally using an order-independent aggregation function. Many techniques from CNNs have analogs in GCNs. Li et. al. [2019] showed that residual connections are crucial for training deep GCNs; we incorporate these into our network design since GCN depth controls how large of a local neighborhood our learned representation for each topological feature can incorporate. A novel feature of graph domains that is pertinent to CAD data is domain heterogeneity. While basic GCNs handle structural heterogeneity of the domain, there is no natural extension of CNNs or RNNs that captures discrete typings of nodes and edges, which may have heterogeneous labels. [Zhang et al. 2018] introduced meta-paths, typed edges connecting all paths with the same node types, and proposed learning a separate message function for each type of meta-path. Learning heterogeneous messages has the disadvantage of greatly expanding model complexity. Our model incorporates both heterogeneous messages and meta-edges, but we leverage the consistent connectivity structure of BREPs to avoid the increase in model complexity usually associated with typed messages by splitting these into a series of homogeneous message passing layers.

3 OVERVIEW

Modeling Mates. Our predictions are generated by scoring and ranking a collection of potential mates. We model a mate as a pair of *mating coordinate frames* (MCFs), a coordinate frame defined for each mated part in reference to their topological entities (such as faces or edges), and a constraint on their relative position and orientation. For example, the two halves of a hinge in Figure 3 are mated by MCFs defined in reference to cylindrical faces. The pair of MCFs, one on each part, define the *mate location*, assembling the parts by aligning the frames. The constraint between MCFs is specified as one of eight *mate types* (Figure 4), which specify which axes the frames can be translated or rotated about relative to one another. For example, Figure 3 shows how a hinge is modeled by a *revolute* mate that disallows translation of the parts, but allow rotation around their common z-axis, letting the hinge swing open and shut.

Each MCF is defined as a tuple of two topological entities from the associated part. One referenced topology defines the origin of the MCF. Some topological entities have more than one location to anchor an MCF to (such as the top, bottom, or middle of a cylinder), so this reference has an associated *origin type* to specify where on the entity to place the origin.¹ The other referenced topological

¹A complete list of origin types is given in Table 4.

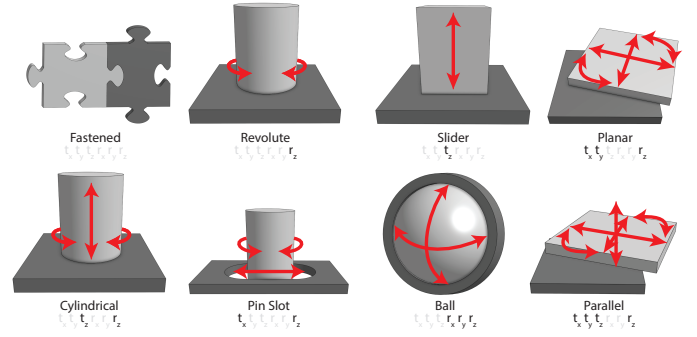


Fig. 4. The 8 mate types. Red arrows illustrate the available degrees of freedom for each mate type.

entity determines the frame's orientation by specifying its z-axis, for example, normal to a referenced planar face. The other coordinate axes are derived from the frame each part was modeled in. See Appendix A for specifics.

This model is simplified from the raw dataset. We discarded a small number of mates that were defined with additional rotational and translational offsets to their MCFs. We also ignore optional per degree-of-freedom range limits, as these are also rare. The raw data we collected also allowed for the MCFs in a mate to be reoriented by one of 8 orthogonal rotations, and for one of the same 8 rotations to be applied to the mate as a whole, giving 512 possible combinations. We canonicalized these down to 8 unique variations for our dataset, and for the models in this paper we limit ourselves to the most common *canonical orientation*.

Interactive Recommendation System. Our method extends a CAD system to provide autocomplete-like suggestions for mates. Automatically suggesting mates is a balance between automation and control from the user. The problem of deciding how to mate BREPs is inherently ambiguous; it depends on the user's design intent, so there is no ground truth. We decided to structure our tool as a recommendation system. In our system, the user clicks on two parts to select them for mating. We use the face that the user clicked as an indication of design intent; the origin of each MCF will be in the neighborhood of the selected faces. We accomplish this by using the selected face as the orientation reference for each MCF, and restrict our search to MCFs with origin on that face or one of its boundaries. This both captures the user's intent and reduces the cost of the search, and is well supported by our dataset; almost all of the origin references in our dataset are within the boundary of the corresponding orientation topological element. Given this input; two BREPs and face selections on each, our system scores and ranks all pairs of MCFs neighboring the selection, and reports the top 6 to the user. Given a particular MCF pair, our system can also predict the most likely mate types. Figure 5 shows our interface for the simple case of two building blocks. The user has selected the top and bottom face of each block to be mated within the system's normal assembly UI. Whenever two faces are selected, our system presents 6 mate location options that the user can choose between with a hotkey, or ignore to continue assembly manually without the

图网络表示学习。通用的基于图的学习技术已广泛应用于几何数据、零件装配结构、社交网络等多种类型的数据领域。本研究通过将这些技术应用于具有BREP文件结构特征的图结构中，实现了对CAD数据的有效处理。关于该领域的深入综述，读者可参阅两篇近期发表的专题论文。[Wu 埃特；埃特 al. 2020a], [张] 埃特；埃特 al. 2020] 本文仅引用最相关的研究。大多数现代研究均遵循Kipf和Welling[2017]提出的图卷积方法（GCN），该方法通过两步消息传递过程将欧几里得卷积推广至图域：首先收集相邻节点的特征作为计算消息，随后利用一种与排序无关的聚合函数对这些消息进行局部聚合。CNN中的许多技术在GCNs中都有对应方法。Li等人。[2019] 研究表明，残差连接对于深度GCN的训练至关重要；我们将这些机制融入网络设计中，因为 GCN 深度控制着我们为每个拓扑特征学习到的表示所能涵盖的局部邻域范围。图域中一个与CAD数据相关的新型特性是域异质性——虽然基础GCN能够处理域的结构异质性，但CNN或RNN并无天然的扩展方式来捕捉节点和边的离散类型，这些节点和边可能具有异质标签。[Zhang等人2018] 提出了元路径概念——即通过带类型标注的边将所有路径连接起来，并建议为每种元路径学习独立的消息函数。学习异质消息存在显著增加模型复杂度的缺点。我们的模型同时整合了异质消息与元边，但通过将这些消息拆分为一系列同质消息传递层，利用BREPs的一致连通结构，从而避免了通常与带类型消息相关的模型复杂度上升问题。

3 概述

配对建模。我们的预测是通过通过对一组潜在配对对象进行评分和排序而生成的。我们把配对对象建模为一对**配对坐标系（MCFs）**——每个配对部件都对应一个基于其拓扑实体（如面或边）定义的坐标系，以及对其相对位置和方向的约束条件。例如，图3中铰链的两个半部分通过基于圆柱面定义的MCFs进行配对；这对MCFs（分别位于每个部件上）共同定义了**配对位置**，通过对齐这些坐标系来组装部件。MCFs之间的约束关系由八种**配对类型**之一来指定（图4），这些类型明确了各坐标系之间可沿哪些轴进行平移或旋转。例如，图3展示了如何通过**旋转配对**来建模铰链——该配对方式不允许部件间的平移，但允许它们绕其共同的z轴旋转，从而使铰链能够开合。

每个MCF被定义为来自相关部件的两个拓扑实体组成的元组。其中一个引用的拓扑结构决定了MCF的原点位置。**某些拓扑实体具有多个可作为MCF锚定点的位置（例如圆柱体的顶部、底部或中部），因此该引用关系会关联一个原点类型以指定在实体上的具体位置。**¹其他被引用的拓扑

¹完整的起源类型列表见表4。

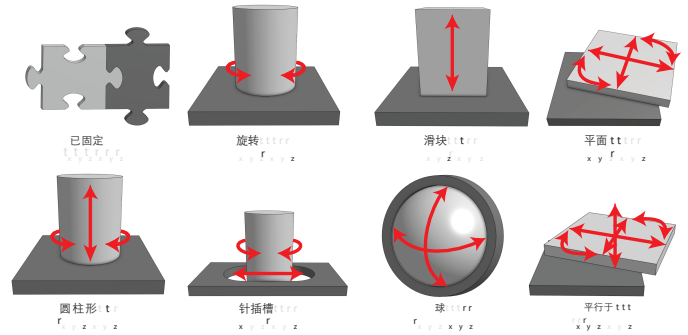


图4：8种配对类型。红色箭头展示了每种配对类型的可用自由度。

实体通过指定其z轴（例如，与参考平面面的法线方向）来确定坐标系的方向。其他坐标轴则根据各部件建模所用的坐标系推导得出。具体细节参见附录A。

该模型是从原始数据集中简化而来。我们剔除了少数那些其运动中心轨迹（MCF）存在额外旋转和平移偏移量的配对结构，并且由于这些情况较为罕见，我们也未考虑可选的自由度范围限制。采集的原始数据还允许对某个配对结构中的运动中心轨迹进行8种正交旋转之一的重新定向，并可对整个配对结构应用这8种旋转中的任意一种，共产生512种可能组合。我们将这些组合归一化为数据集中的8种独特变体；在本文构建的模型中，我们仅采用最常见的**标准朝向**。

交互式推荐系统。我们的方法扩展了CAD系统，可为配对对象提供类似自动补全的建议。自动推荐配对对象需要在自动化与用户控制之间取得平衡。关于如何对BREP进行配对的问题本身具有模糊性；它取决于用户的设计意图，因此不存在“基准真值”。我们决定将该工具构建为一个推荐系统：在本系统中，用户点击两个部件以选择它们进行配对；我们利用用户点击的位置作为设计意图的指示——每个MCF的原点都将位于所选面的邻域内。我们通过将所选面作为每个MCF的定向参考，并将搜索范围限定于原点位于该面或其边界上的MCF来实现这一目标。这种方法既能准确捕捉用户的意图，又能降低搜索成本，且得到了我们数据集的有力支持——数据集中几乎所有的原点参考均位于相应定向拓扑元素的边界之内。基于上述输入——即两个BREP模型及其各自的面选择——本系统会计算并排序所有与该选择相邻的MCF对，并向用户报告排名前6的候选结果。对于任意特定的MCF对，本系统还可预测最可能的配对类型。图5展示了针对两个构建模块这一简单场景的系统界面：用户可在系统的标准装配UI中，选择每个模块的顶部和底部面以进行配对。每当选择两个面时，本系统将呈现6种配对位置选项供用户选择（可通过快捷键操作），或选择忽略这些选项，从而继续手动进行装配，而无需受自动补全功能的干扰。若用户选择某一配对建议，系统将在部件之间插入一个

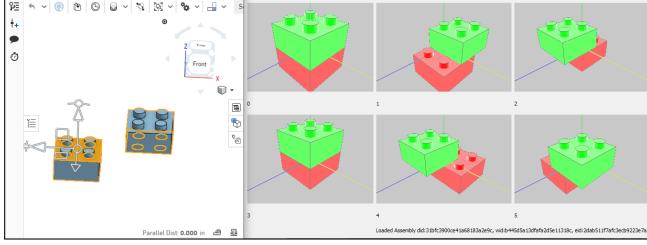


Fig. 5. Our Onshape Extension. Left: Onshape assembly with two parts selected. Right: Top 6 predicted mate locations. The first and fourth suggestions look similar because differently defined mates can resolve to the same location; the first option is centered on middle of the blocks while the fourth is centered on the front peg.

autocomplete impeding their workflow. If the user selects a mate suggestion, a *fastened* mate is inserted between the parts, and the user can then select a mate type. Our system also ranks the mate type likelihood conditioned on the chosen mate location.

Predicting Mates. We score and classify mates using SB-GCN, our graph neural network for BREPs. This message passage network is applied to each input BREP to produce representations for each topological entity that encode the local neighborhood, as well as a global descriptor for each part. Each MCF is encoded by the representations of each referenced entity, the origin type, and the global part descriptor, then we train a classifier to score each MCF (the *mate location task*), or classify its mate type (the *mate type task*). We train this model over all compatible mates from our dataset. Separating the BREP representation learning from the mate classification model is important because while each CAD system has its own constraint model for mates, most use BREPs as their underlying shape representation and construct their mate by reference to topological entities. By explicitly learning to represent each type of topological entity, our system can be adapted to integrate into other CAD systems.

4 DATASET

We scraped all publicly available documents from Onshape using their API. We collected 255,213 assemblies with at least one mate, totaling 3,989,164 mates. However, a significant number of these are complete or partial duplicates created by reuse of common hierarchical sub-assemblies, and by copying and modification of assemblies in this public repository.

To clean this dataset, we first expanded and then flattened all sub-assembly references, and filtered down to only top level assemblies. We also removed all incomplete mates, and pairs of parts with multiple mates between them. To identify identical parts and assemblies, we created a BREP fingerprint from the number of each type of topological entity and the moment of inertia tensor and center of mass of the represented solid. We used this to deduplicate re-used parts, and to identify duplicate mates by MCFs, mate types, and mated parts. Assemblies were deduplicated by their mates. The dataset contains 92,529 unique top level assemblies. These unique assemblies range in size from 1 to 13183 mates, with an average of

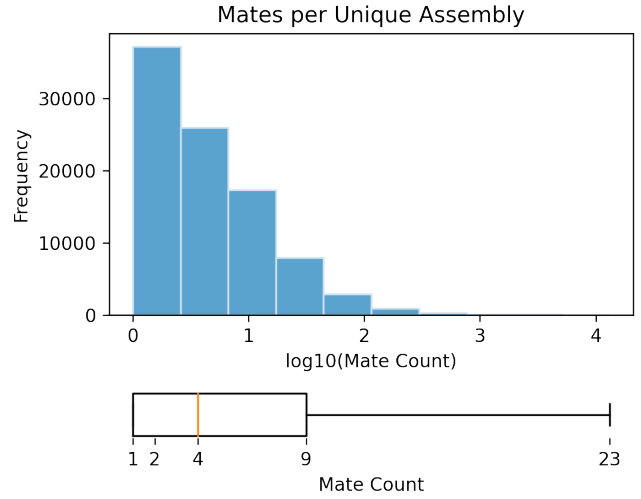


Fig. 6. Number of mates per assembly in the dataset. Top: histogram of log assembly size. Bottom: Box-plot of assembly sizes with 10th, 25th, 50th, 75th, and 90th percentiles labeled.

Table 1. Mate type frequencies.

Mate Type	Frequency
FASTENED	62.1%
REVOLUTE	12.7%
PLANAR	11.8%
SLIDER	5.3%
CYLINDRICAL	5.1%
PARALLEL	1.8%
BALL	0.6%
PIN SLOT	0.5%

12, and a 90th percentile of 23. Figure 6 shows the distribution of mates per assembly.

At the mate level, we have 541,635 unique mates. Table 1 gives the frequency of each mate type among unique mates. Our dataset contains a total of 376,362 unique BREP models used in mates.

5 METHODS

Graph Representation Learning for BREPs. BREPs encode both the geometry and the topology of a shape as a collection of topological entities with associated parametric geometry. Faces, edges, and vertices are topological entities associated with parametric geometry functions of surfaces (2D), curves (1D), and points (0D). While the *parameters* (radius, normal, half-angle; see Appendix A for a full list) of these functions are defined for each topological entity, the domain on which to evaluate each function is not. This information is encoded by the *topological* structure of the BREP, defining relationships between topological entities. Lower dimensional entities are related to higher dimensional neighbors by a *boundary of* relationship, and their geometry implicitly defines the domain bounds of these neighbors. A fourth type of topological entity, loops, consist of closed paths of edges. Edges are related to loops by a *component*

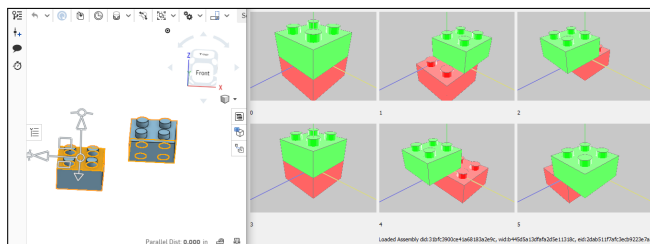


图5. 我们的Onshape扩展插件。左图：Onshape装配体中已选中两个零件；右图：前6个预测的配合位置。第一项与第四项建议看起来颇为相似，这是因为定义不同的配合方式可能得出相同的定位结果；其中，第一种方案的中心点位于块体的中间位置，而第四种方案的中心点则位于前部销钉处。

“已固定”配对，并允许用户随后选择具体的配对类型。此外，本系统还能根据所选配对位置，对各配对类型的出现概率进行排序。

预测配对对象。我们利用SB-GCN——我们为BREPs设计的图神经网络——对配对对象进行评分和分类。该消息传递网络应用于每个输入 BREP，以生成每个拓扑实体的表征（这些表征编码了局部邻域信息）以及每个部件的全局描述符。每个MCF均通过各引用实体的表征、原点类型及全局部件描述符进行编码；随后，我们训练一个分类器来对每个MCF进行评分（即**配对位置任务**），或对其配对类型进行分类（即**配对类型任务**）。我们使用数据集中的所有兼容配对对象来训练该模型。将 BREP 表征学习与配对分类模型分离至关重要，因为尽管每套CAD系统都拥有自身的配对约束模型，但大多数系统均采用BREPs作为其底层形状表示，并通过参考拓扑实体来构建配对关系。通过显式学习表征各类拓扑实体，我们的系统可适配并集成至其他CAD系统中。

4 数据集

我们通过Onshape的API抓取了该平台所有公开可用的文档，共收集到255,213个至少包含一个配对元件的组装体，总计3,989,164个配对元件。然而，其中相当一部分为完全或部分重复项，这些重复项是通过重复使用常见的层级子组装体以及复制和修改该公共存储库中的组装体而产生的。

为清理该数据集，我们首先对所有子装配参考进行了扩展和展平操作，随后筛选出仅包含顶级装配的条目；同时移除了所有不完整的配对部件以及彼此间存在多个配对关系的部件对。为了识别相同的部件和装配体，我们根据各类拓扑实体的数量、所代表实体的转动惯量张量及其质心，生成了 BREP 指纹。利用此指纹，我们实现了对重复使用的部件进行去重，并通过MCFs、配对类型及配对部件来识别重复的配对关系。装配体则依据其配对部件进行去重处理。该数据集共包含92,529个独特的顶级装配体，这些独特装配体的规模从1个到13,183个配对部件不等，平均值为

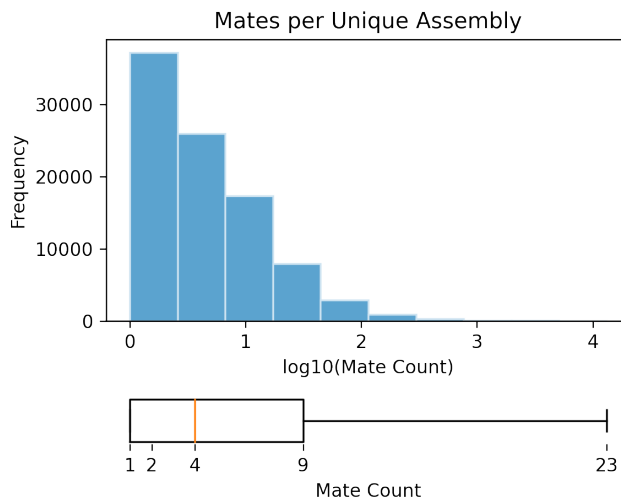


图6. 数据集中每个组装体的配对数量。上图：组装体大小的对数分布直方图；下图：组装体大小的箱线图，并标注了第10、25、50、75和90百分位数。

表 1. 配对类型频率。

茶类	频繁性
系紧	62.1%
旋转测量仪	12.7%
平面型	11.8%
滑块	5.3%
圆柱形	5.1%
并行	1.8%
球	0.6%
PIN插槽	0.5%

12，其第90百分位数为23。图6展示了每个组装结果对应的配对数分布情况。

在配对水平上，我们共有541,635个独特的配对。表1列出了各配对类型在这些独特配对中的出现频率。我们的数据集中共包含376,362个用于配对的唯一 BREP 模型。

5 方法

用于BREP的图表示学习。BREP将形状的几何结构和拓扑结构编码为一组具有关联参数化几何特征的拓扑实体集合。面、边和顶点都是与曲面（二维）、曲线（一维）及点（零维）的参数化几何函数相关的拓扑实体。虽然这些函数的参数（半径、法线、半角；完整列表参见附录A）为每个拓扑实体定义明确，但各函数的计算域却并非如此。这些信息通过 BREP 的拓扑结构进行编码，该结构定义了拓扑实体之间的关系：低维实体通过边界关系与高维邻接体相连，其几何特征隐式地界定了这些邻接体的域范围。第四种拓扑实体——环路由闭合边路径构成；边通过隶属关系与环路关联，并本身也是面的边

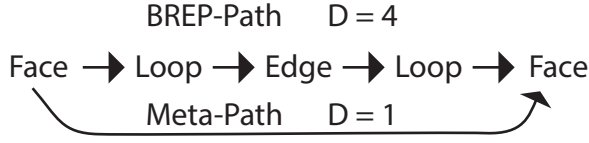


Fig. 7. The BREP graph distance between neighboring faces is 4; with meta-paths, this is reduced to 1, shrinking the graph diameter by about 4 times.

of relation, and are themselves a boundary of a face. Loops have one parameter to specify if they are an inner or outer boundary of a surface. Figure 2 (A) shows a simple BREP with faces bounded by loops composed of edges bounded by vertices.

The *boundary* of and *component* of relations define a directed graph in which the topological entities are *nodes*, pictured in Figure 2 (B). This graph is *heterogeneous* in both its node types and relation types.² Recent and concurrent works [Lambourne et al. 2021; Willis et al. 2020] have used a simplified, *homogeneous* version of this graph with only face nodes to build message passing networks for learning BREP representations. These representations are insufficient for learning how to mate BREPs however, because BREP mates can be defined relative to topological entities of any type. Our representation learning must therefore handle the full, heterogeneous BREP graph.

Introducing additional nodes and relations presents additional challenges to representation learning. Having multiple types of bounds increases the path length between topological faces; for example, geometrically neighboring face nodes are at graph distance four. We would like our learned embeddings to capture boundary information and local structure, so we want a wide receptive field for our convolutions. In graph convolutional networks, the receptive field is a function of the number of convolutional layers, but as [Li et al. 2019] point out, state-of-the-art GCNs are typically only 3-4 layers deep as deeper graph networks are difficult to train. This would barely reach the neighboring face! (See Figure 7.)

We adopt two strategies to overcome this. First, we use residual connections in our network, which [Li et al. 2019] showed help stabilize the training of deep GCNs. We also add undirected meta-paths [Zhang et al. 2018] of (face←loop←edge→loop→face) relations to connect geometrically adjacent faces and reduce the graph diameter as shown in Figure 7. These match the undirected edges of [Willis et al. 2020].

Learning a different message passing function for each type of relation adds significant time and memory complexity to the model, and so limit the number of layers we can add and the size of our representation. The complexity of a BREP neural network is important to control because BREPs range wildly in complexity, so a large and complex part can easily exhaust available GPU memory. BREPs have a very regular connectivity structure that we use to partition the graph into a *Structured BREP Graph* that we can use to get the advantages of a wide receptive field, heterogeneous graph network for less than the cost of an equivalent homogeneous network.

²We use *node* and *relation* to refer to *graph* vertices and edges throughout to avoid confusion with topological entity types of the same names.

Since each type of node only has one type of relation, to exactly one other type of node, we can separate our BREP graphs into four *tiers* by node type, as pictured on the left side of Figure 8. We break our heterogeneous message passing into separate homogeneous message passing layers between adjacent tiers. We first pass messages up through the layers from vertices to faces to give each topological entity its boundary information. Then we perform several layers of message passing over just the face-face meta-paths to widen the receptive field, collecting neighborhood information at each face. Finally, we pass this neighborhood information backwards *down* the tiers from faces to vertices, so in the end, all topological entities have incorporated information about their boundaries and their local neighborhood.

Splitting the heterogeneous convolution into several smaller homogeneous ones saves a significant amount of memory (one sixth the number of parameters) without losing any information on the upward or downward passes. Messages passed from, for example, a loop to a face, can only contain incrementally more information until the information from the vertices at the bottom of the structure reaches the loop anyway, so it is more efficient to incrementally accrue that information in a single upward pass. We call our tiered graph network a *Structured BREP Graph Convolutional Network* (SB-GCN).

SB-GCN. Formally, we define a Structured BREP Graph G as a collection of node sets, $N = F \cup L \cup E \cup V$, where F , L , E , and V are the face, loop, edge, and vertex nodes respectively. These nodes are then connected by directed, bipartite relation sets $V : E$, $E : L$, and $L : F$ and their transposes $F : L$, $L : E$, and $E : V$. For, say, $V : E$ the set contains relations r_{uv} that connects the nodes $u \in V$ to $v \in E$. There is also a set of undirected relations $F : F$ which are the meta-path relations between geometrically adjacent faces, computed by vertex-elimination on non-Face nodes in the graph. SB-GCN takes G and (possibly heterogeneous) input feature vectors on the nodes of G as $N^{(in)} = V^{(in)} \cup E^{(in)} \cup L^{(in)} \cup F^{(in)}$ and produces output feature matrices $V^{(out)}$, $E^{(out)}$, $L^{(out)}$, and $F^{(out)}$ via a series of structured graph convolutions. Figure 8 shows the network architecture.

First a shared MLP transforms all input feature vectors into a common feature space prior to convolution:

$$N^{(0)} = \text{MLP}^{(in)} \left(N^{(in)} \right). \quad (1)$$

Next a cascade of residual graph convolutions are computed. We adapt the Max-Relative GCN (MRGCN) that [Li et al. 2019] found effective for training deep GCNs with residual connections:

$$\begin{aligned} \Sigma_v^{(l)} &= \max \left(\left\{ h_u^{(l)} - h_v^{(l)} \mid u \in \mathcal{N}_{\tau(u) : \tau(v)}(v) \right\} \right) \\ h_v^{(l+1)} &= h_v^{(l)} + \text{MLP}^{(l)} \left(\text{cat} \left(h_v^{(l)}, \Sigma_v^{(l)} \right) \right). \end{aligned} \quad (2)$$

This is computed for all $h_v^{(l)} \in N^{(l)}$ where $h_v^{(l)}$ is the feature of the node $v \in N^{(l)}$. $\tau(v)$ is an operator for $v \in N^{(l)}$ which returns which of F , L , E , or V that v is in, that is the topological type of the node v , and so $\mathcal{N}_{\tau(u) : \tau(v)}(v)$ is the neighborhood of v using the relation set $\tau(u) : \tau(v)$. $\text{MLP}^{(l)}$ is then the l th layer's MLP of a linear layer, batch normalization, and ReLU. The important difference in

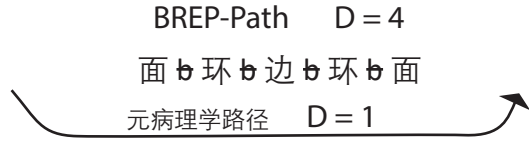


图7. 相邻面之间的 BREP 图距离为4; 引入元路径后, 该距离降至1, 使图直径缩小约4倍。

界。环路仅需一个参数即可指定其作为曲面的内边界或外边界。图2(A)展示了一个简单的 BREP, 其面由由顶点围成的边构成的环形结构所界定。

边界关系与组件关系共同定义了一个有向图, 其中拓扑实体作为节点存在 (如图2(B)所示)。该图在节点类型和关系类型上均具有 *异构性*。² 近期及同期的研究[Lambourne 等人 2021; Willis 等人 2020]采用仅包含面节点的简化版 *同构图* 结构来构建用于学习 BREP 表示的消息传递网络。然而这些表示方法不足以实现 BREP 配对关系的学习, 因为 BREP 配对可相对于任何类型的拓扑实体进行定义。因此我们的表示学习必须处理完整的异构 BREP 图结构。

引入额外的节点和关系会给表示学习带来新的挑战。多种类型的边界划分会增加拓扑面之间的路径长度; 例如, 几何相邻的面节点之间的图距离可达四。我们希望所学习到的嵌入能够捕捉边界信息和局部结构, 因此需要为卷积层设置宽广的感受野。在图卷积网络中, 感受野的大小取决于卷积层数; 但正如 [Li 等 2019] 所指出的, 最先进的 GCN 通常仅具有 3-4 层深度, 因为更深的图网络难以训练。这种结构几乎无法覆盖相邻面的信息! (参见图7。)

我们采用两种策略来解决这一问题。首先, 在网络中使用残差连接 [Li 等人 2019], 这有助于稳定深度 GCN 的训练; 同时我们还添加了 (面 $\leftarrow$ 环 $\leftarrow$ 边 $\leftarrow$ 环 $\leftarrow$ 面) 关系的无向元路径 [Zhang 等人 2018], 用于连接几何相邻的面并减小图直径 (如图7所示), 这些设计与 [Willis 等人 2020] 提出的无向边概念一致。

为每种关系类型学习不同的消息传递函数会显著增加模型的时间和内存复杂度, 从而限制可添加的层数量及表示结构的规模。BREP 神经网络的复杂度需要严格控制, 因为 BREPs 的复杂程度差异极大; 过于庞大复杂的部分极易耗尽 GPU 可用内存。BREPs 具有非常规整的连接结构, 我们利用这一特性将图划分为 *结构化 BREP 图*, 从而以低于等效同质网络的成本, 获得宽感受野和异构图网络带来的优势。

² 为避免与同名的拓扑实体类型产生混淆, 我们在全文中均使用 *节点* 和 *关系* 来指代图中的顶点和边。

由于每种节点类型仅与一种关系类型相关联, 且该关系仅指向另一种特定类型的节点, 我们可以根据节点类型将我们的 BREP 图划分为四个层级, 如图8左侧所示。我们将异构消息传递分解为相邻层级之间独立的同构消息传递层。首先, 我们通过这些层将消息从顶点传递至面, 从而为每个拓扑实体提供其边界信息。接着, 我们在仅限于面-面元路径上进行多层消息传递, 以扩大感受野, 并在每个面处收集邻域信息。最后, 我们将这些邻域信息沿层级 *向下* 从面传递至顶点, 最终使得所有拓扑实体均能整合其边界及局部邻域的信息。

将异构卷积分解为多个较小的同构卷积单元, 可以在不损失上下传递过程信息的前提下显著节省内存 (仅需参数数量的六分之一)。例如从循环节点传递至面部节点的消息只能逐步增加信息量——因为结构底部顶点的信息最终总会传到循环节点; 因此, 在单次向上传递过程中逐步累积这些信息更为高效。我们将这种分层图网络称为 *结构化 BREP 图卷积网络 (SBGCN)*。

SB-GCN。形式上, 我们定义一个结构化 BREP 图 G 为一组节点集的集合: $N = F \cup L \cup E \cup V$, 其中 F 、 L 、 E 和 V 分别表示面节点、环节点、边节点和顶点节点。这些节点通过有向二分关系集 $V:E$ 、 $E:L$ 以及 $L:F$ 及其转置关系 $F:L$ 、 $L:E$ 和 $E:V$ 进行连接。例如, 对于 $V:E$, 该集合包含连接节点 $u \in V$ to $v \in E$ 的关系 r_{uv} ; 此外还存在一组无向关系 $F:F$, 它们是几何相邻面之间的元路径关系, 通过在图中非面节点上执行顶点消除运算计算得出。SB-GCN 接收 G 以及 G 节点上的 (可能为异构的) 输入特征向量, 将其表示为 $N(i) = V(i) \cup E(i) \cup L(i) \cup F(i)$, 并通过一系列结构化图卷积操作生成输出特征矩阵 $V(out)$ 、 $E(out)$ 、 $L(out)$ 及 $F(out)$ 。图8展示了该网络架构。

首先, 一个共享 MLP 会在卷积操作前将所有输入特征向量转换至同一特征空间:

$$N(i) = \text{MLP}(i) \cdot N(i)'. \quad (1)$$

接下来, 计算一系列残差图卷积操作。我们采用了 [Li 等人 2019] 在训练带有残差连接的深度 GCN 时发现有效的 Max-Relative GCN (MRGCN) 方法:

$$\begin{aligned} \Sigma_v^{(l)} &= \max \left(\left\{ h_u^{(l)} - h_v^{(l)} \mid u \in \mathcal{N}_{\tau(u): \tau(v)}(v) \right\} \right) \\ h_v^{(l+1)} &= h_v^{(l)} + \text{MLP}^{(l)} \left(\text{cat} \left(h_v^{(l)}, \Sigma_v^{(l)} \right) \right). \end{aligned} \quad (2)$$

此计算适用于所有 $\hat{v} \in N^{(l)}$, 其中 $\hat{v} \in N^{(l)}$ 表示节点 $v \in N^{(l)}$ 的特征。 $\mathcal{E}(v)$ 是针对 $v \in N^{(l)}$ 的一个算子, 用于返回 F 、 L 、 E 或 V 中 v 所属的类别——即节点 v 的拓扑类型; 因此, $\mathcal{N}_{\mathcal{E}(v)}(v)$ 表示利用关系集 $\mathcal{E}(v)$ 生成的 v 的邻域。 $\text{MLP}^{(l)}$ 则是第 l 层由线性层、批归一化和 ReLU 组合而成的 MLP。两者之间的关键区别在于

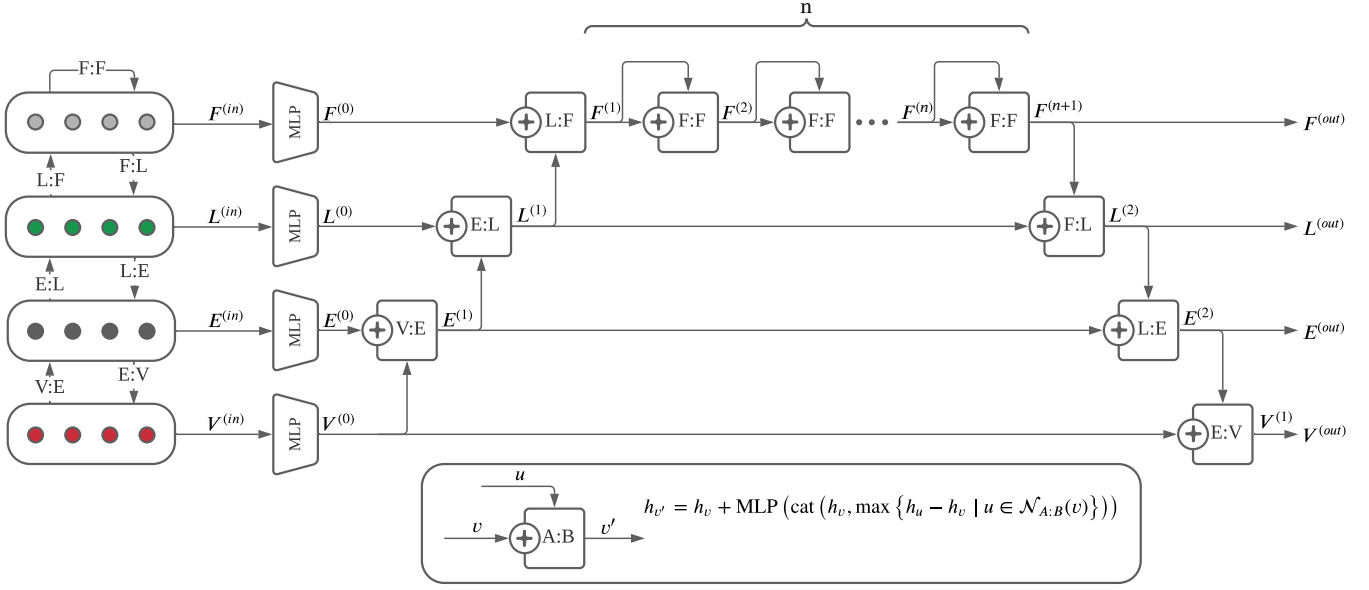


Fig. 8. The SB-GCN architecture. BREP graph nodes are structured into 4 tiers by their topological entity type, with boundaries below the type they bound. Boundary information is passed up the tiers from vertices through to faces via residual MRGCN convolutions. Additional undirected MRGCN convolutions between faces quickly widen the network's receptive field, then the aggregated neighborhood information is propagated back down the tiers from faces through to vertices via more convolutions. Separating the convolutions into layers allows each convolutional layer to be homogeneous while still learning separate message passing functions for each type of graph relation.

our convolution is that the adjacency of relations is determined depending on the layer, denoted by the subscript of the neighborhood operator. For the first three layers, relations are in the order of the BREP hierarchy: Vertex to Edge, Edge to Loop, and Loop to Face. Then, the Face-to-Face meta-relations are used for the inner k layers. Finally, for the last three layers, the relations are reversed from the original BREP relations: Face to Loop, Loop to Edge, and Edge to Vertex.

Input Features. We use both the analytic geometry of each BREP topological element and a few computed geometric summary features as the per node input feature vectors for SB-GCN. The analytic features are a one-hot encoding of the parametric geometry function associated with each node, plus a vector of each function's parameters. We exclude parameters for functions without a fixed parameter list size (such as B-Splines and NURBS surfaces) as well as geometry defined in reference to other geometry (like spun surfaces or intersection curves). See Appendix A for a full list parametric functions and their parameters. These more complex topological elements are never referenced for MCFs, so excluding their parameters is not too detrimental. We also exclude the origin parameter because we found that in practice it is frequently far away from the realized geometry once boundaries are taken into account and acted like noise.

To compensate for this lost localization feature, and give useful summarizing information about the more complex surfaces we exclude parameters for, we compute 4 additional geometric summaries. To localize each node in space, we compute its center of

mass and an axis-aligned bounding box. We know that the size of each topological entity is a critical feature to determine which fit together, so we explicitly compute surface areas and arc lengths and add those features. Finally, we compute the moment-of-inertia tensor to summarize the geometry of more complex topological entities. We tested adding PointNet derived features to each entity, but found these to be unhelpful given the very low sample count per topological entity, and so do not include them.

Prior to extracting analytic and geometric features, we normalize the part geometry by translating both parts to the origins of their respective coordinate systems, and applying a consistent scale factor to both so the largest dimension of either part has unit length.

Predicting Mates with SB-GCN. We apply SB-GCN to solve two mating tasks: *location prediction*, which scores and ranks pairs of MCFs adjacent to selected faces on two parts, and *mate type prediction*, which classifies the mate type for a mate with a specific pair of MCFs. We use the same classification network with a different output head for both tasks, shown in Figure 9. Our classifier uses SB-GCN in a siamese configuration over both input BREPs to learn an embedding for each topological entity. A global part feature is computed for each part with a meanpool of these embeddings. We then construct a feature vector for each MCF in the mate by concatenating the embeddings of the MCF's orientation and origin topological entities, a one-hot encoding of the origin type, and the global part feature. These MCF features are concatenated and fed to an output MLP for either location prediction (one output) or mate type classification (8 outputs).

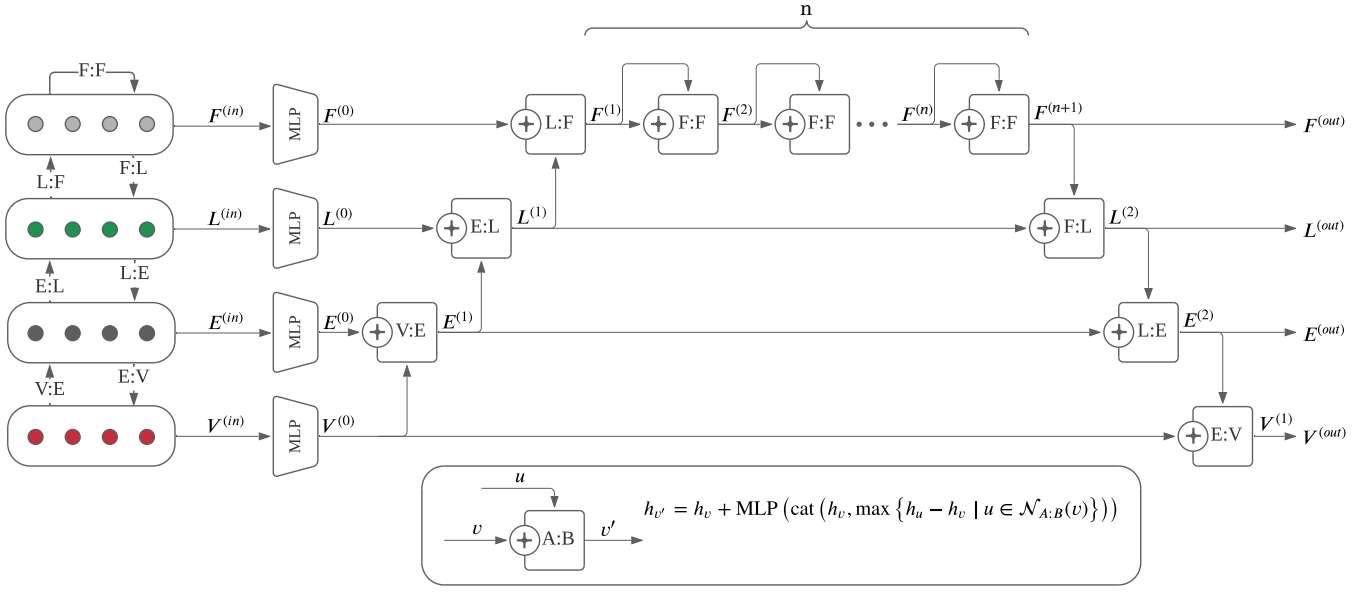


图8. SB- GCN 架构。BREP图节点根据其拓扑实体类型被划分为4个层级，边界信息位于其所关联类型的下一层级。边界信息通过残差MRGCN卷积从顶点向上传递至面层。面层之间的额外无向MRGCN卷积可快速扩展网络的感受野；随后聚合后的邻域信息再通过更多卷积从面层向下传递至顶点。将卷积操作分层处理可使各卷积层保持同质性，同时为每种图关系类型学习独立的消息传递函数。

我们的卷积操作在于：关系的邻接关系取决于各层，这由邻域算子的下标表示。对于前三层层，关系遵循BREP层级结构：顶点至边、边至环、环至面。随后，对于内部 k 层，采用面至面的元关系。最后，对于后三层层，关系顺序与原始BREP关系相反：面至环、环至边、边至顶点。

输入特征。我们同时使用每个BREP拓扑元素的解析几何特征以及若干计算得出的几何汇总特征作为SB-GCN算法中每个节点的输入特征向量。解析特征包括与每个节点相关的参数化几何函数的一热编码，以及该函数各参数组成的向量。对于参数列表大小不固定的函数（如B样条和NURBS曲面）以及基于其他几何体定义的几何结构（如旋转曲面或交线曲线），我们均排除其参数；完整的参数化函数及其参数列表详见附录A。这些更为复杂的拓扑元素在MCF模型中从未被引用，因此排除其参数并不会造成显著影响。此外，我们还排除了原点参数，因为实践表明在考虑边界条件后，该参数通常远离实际几何形状，并会像噪声一样产生干扰。

为弥补这一定位功能的缺失，并针对我们排除参数范围内的复杂表面提供有效的汇总信息，我们额外计算了四项几何特征。为在空间中精确定位每个节点，我们计算其质心并生成轴

对齐的边界框。鉴于各拓扑实体的尺寸是决定其能否成功匹配的关键特征，我们明确计算了表面积和弧长并将这些数据纳入分析。最后，我们通过计算惯性矩张量来概括复杂拓扑实体的几何结构。虽然我们尝试将PointNet衍生特征应用于每个实体，但由于每个拓扑实体的样本数量极少，这些特征并未显现显著价值，因此最终未将其纳入分析。

在提取分析特征和几何特征之前，我们通过将两个部件均平移至各自坐标系的原点，并对两者应用统一的比例因子来标准化部件几何形状，从而使任一部件的最大尺寸具有单位长度。

利用SB-GCN预测配对对象。我们应用SB-GCN来解决两个配对任务：**位置预测**——对两个部件上与选定面相邻的MCF对进行评分和排序；以及**配对类型预测**——为特定MCF对对应的配对对象分类其配对类型。我们采用相同的分类网络，但为这两个任务配置不同的输出头，如图9所示。我们的分类器在两个输入BREP上采用双胞胎结构配置，通过SB-GCN学习每个拓扑实体的嵌入表示。针对每个部件，我们通过这些嵌入向量的均值池化计算出全局部件特征。随后，我们通过拼接MCF的定向与原点拓扑实体的嵌入向量、原点类型的独热编码以及全局部件特征，为配对中的每个MCF构建特征向量。这些MCF特征被拼接后输入到输出MLP层，用于位置预测（单个输出）或配对类型分类（8个输出）。

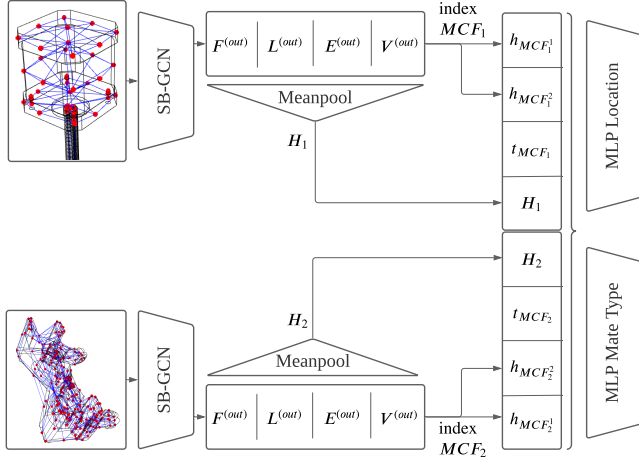


Fig. 9. Our classification network. SB-GCN is used to learn embeddings for every topological entity of each part. The embeddings of the topological entities defining each MCF's origin and orientation (written as the topological entities MCF_i^1 and MCF_i^2 respectively, and thus with embeddings $h_{MCF_i^1}$ and $h_{MCF_i^2}$) are concatenated with a one-hot encoding of the MCF's *origin type*, t_{MCF_i} , and a global part feature, H_i , computed via a meanpool, to construct MCF features. Each parts' MCF features are concatenated and then classified or scored with an output MLP, depending on the task.

Dataset. Our training data is derived from an earlier version of the dataset described in Section 4, which has similar statistics, but fewer overall mates. We also applied further filters; we removed mates with overly simple or overly complex geometry (measured by face count) to weed out basic primitives and remove large part outliers that could overwhelm GPU memory.

We also augmented our dataset by finding alternative ways to construct equivalent MCFs. Figure 10 illustrates several ways of constructing the same MCF for a simple part using two different orientation topology selections, as well as other MCFs that can be constructed from those face selections. We augment our dataset by adding mates using all equivalent MCFs as positive examples for mate location, and include MCFs that use the same orientation topology but are not equivalent to construct negative examples. These are grouped by selected topology on each part to create selection examples we train and test on for location. For mate type, we only use the augmented positive examples. Since our UI only supports selecting faces, we remove selection examples that aren't based on a face selection. We apply one final filter to remove MCFs which are not canonically oriented, as described in Section 3, and selection examples with over 10,000 potential mates to limit per-batch memory requirements. After augmentation and filtering we have 267,385 selection examples to train over.

Training. All models were optimized using the Adam optimizer with a learning rate of 0.001 and betas of (0.9, 0.999). For mate location, we computed an augmented true positive set as described above, labeled as positive examples, and for each compatible potential user selection, all other compatible mates labeled as negative examples. We trained our output classifier with a binary cross-entropy

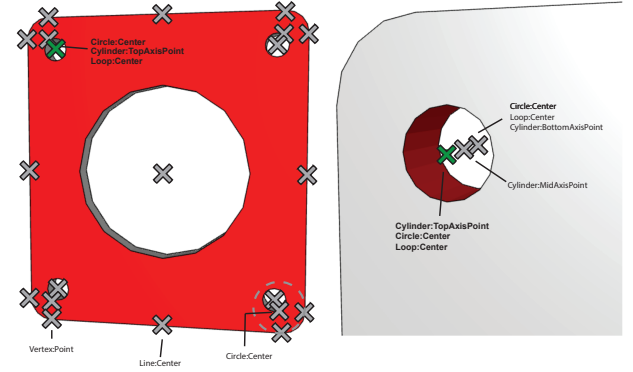


Fig. 10. MCF data augmentation. The red face is the selected orientation topology. Green Xs are MCFs equivalent to the ground truth, and grey Xs are other MCFs that can be constructed from those selections, which are used to construct negative location examples. Selected MCFs have been annotated with their location topology and origin types. Note that there are multiple locations aliasing the ground truth MCFs for both selections.

loss, weighted batch-wise. For mate type, we similarly computed an augmented set of true positive mates, labeled with the ground truth mate type (since we are conditioning this model on the mate location), and then trained with cross-entropy loss. Model selection was performed using the average of the hit ratio at k : over $k \in [1, 10]$ for location and over $k \in [1, 8]$ for mate type.

Implementation. We use Parasolid [Siemens AG [n.d.]] to parse BREP files, which we convert to graphs as described above. Our GCN is implemented in Pytorch Geometric [Fey and Lenssen 2019]. We implemented our onshape integration as a browser extension that captures the user's inputs to Onshape, and uses the Onshape API to download BREPs and assembly definitions and to create mates. For SB-GCN we used $n = 6$ inner layers, and 64 neurons for every linear layer. 64 neurons were used again for the final predictor MLPs' linear layer.

6 RESULTS

Tasks and Metrics. We evaluated our system's performance in two tasks; predicting the exact mate location conditioned on the user's part and face selections, and predicting the type of a mate conditioned on its ground truth location. Our system is structured as a recommendation system, so as our primary evaluation metric we use hit ratio at k ; the likelihood of presenting a correct solution to the user at various depths (k) of recommendation list. For the mate location problem in particular, our system as-implemented presents 6 suggestions to the user, so we use this threshold to define our overall system accuracy. We also define the NDCG* as the average inverse log rank of the first correct suggestion to compare methods with a single number. This is a lower-bound approximation to the Normalized Discounted Cumulative Gain commonly used to evaluate retrieval systems, differing in that we don't award additional gain for correct predictions past the first.

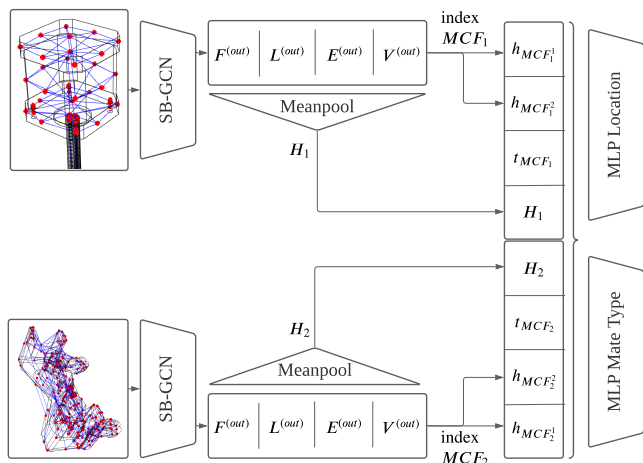


图9. 我们的分类网络。SB- GCN 用于学习每个部件中每个拓扑实体的嵌入向量。定义每个MCF的起源与方向的拓扑实体（分别记为拓扑实体 $MCF1i$ 和 $MCF2i$ ，其对应的嵌入向量为 $\hat{f}_{MCF1i}$ ）并将 $\hat{f}_{MCF2i}$ 与MCF的来源的一热编码进行拼接类型 t_{MCFi} 以及通过均值池化计算得到的全局部件特征 H_i ，用于构建MCF特征。各部件的MCF特征被拼接后，根据具体任务需求，通过输出MLP进行分类或评分。

数据集。我们的训练数据源自第4节所述数据集的早期版本，该版本具有相似的统计特征，但总体配对数量较少。我们还应用了额外过滤条件：剔除了几何结构过于简单或过于复杂的配对（通过面数衡量），以排除基础几何体，并移除可能占用过多GPU内存的大尺寸异常数据。

我们还通过寻找构建等效MCF的替代方法来扩充数据集。图10展示了使用两种不同的定向拓扑选择方式，为简单零件构建相同MCF的多种方法，以及可从这些面选择生成的其他MCF。我们通过添加采用所有等效MCF作为配对定位正例的配对实例，并纳入采用相同定向拓扑但并非等效的MCF作为负例，从而扩充了数据集。这些样本按各零件所选拓扑进行分组，以生成用于位置定位训练与测试的选择示例。对于配对类型，我们仅使用扩充后的正例；由于用户界面仅支持面选择功能，我们剔除了非基于面选择的选型示例；同时，如第3节所述，我们应用最终过滤步骤以移除未采用标准定向的MCF，以及包含超过10,000个潜在配对的选型示例，以限制单批训练的内存需求。经过数据增强与过滤处理后，我们获得了267,385个可用于训练的选择示例。

训练。所有模型均采用Adam优化器进行优化，学习率为0.001， ζ 参数取值为(0.9,0.999)。对于配对位置，我们按照上述方法计算了一个增强的真实正例集合，并将其标记为正样本；而对于每个兼容的潜在用户选择，其余所有兼容的配对对象则被标记为负样本。我们使用二元交叉熵损失函数，按批次加权的方式

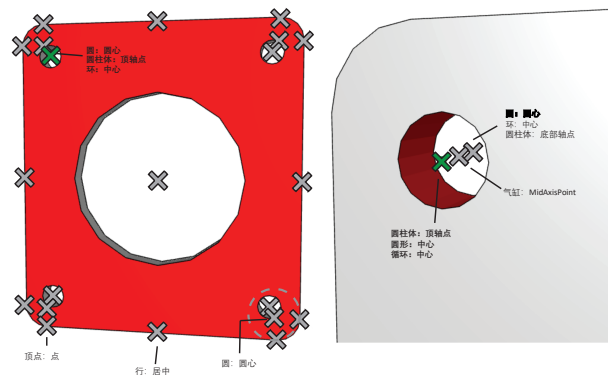


图10. MCF数据增强。红色面表示所选方向拓扑；绿色X点代表与真实标注一致的MCFs；灰色X点则表示可从上述选择中构建的其他MCFs，这些MCFs用于生成样本位置示例。已为所选MCFs标注其位置拓扑和起源类型。需注意的是，对于两种选择情况，均存在多个与真实标注MCFs对应的位置别名。

对输出分类器进行训练。对于配对类型，我们同样计算了一个增强的真实正例配对集合——这些配对对象均标注了真实配对类型（由于我们是基于配对位置来训练该模型），随后使用交叉熵损失函数进行训练。模型选择过程采用 k 值下的命中率平均值： $k \in [1, 10]$ for location and over $k \in [1, 8]$ ，用于配对类型的评估。

实现方法。我们使用Parasolid[Siemens AG[未注明日期]]解析BREP文件，并将其转换为上述所述的图结构。Our GCN 模块基于Pytorch Geometric [Fey与Lenssen 2019]实现。我们的Onshape集成功能通过浏览器扩展程序实现：该程序捕获用户对Onshape的输入数据，并利用Onshape API下载BREP文件和装配定义信息以及创建配合体。对于SB- GCN 模型，我们采用 $n=6$ 个内层结构，每个线性层配置64个神经元；最终预测器MLP模型的线性层同样使用了64个神经元。

6 结果

任务与评估指标。我们通过两项任务评估了系统的性能：一是根据用户的部位和面部选择预测精确的配对位置；二是根据配对对象的真实位置预测其类型。我们的系统采用推荐系统架构，因此将 k 位上的命中率作为主要评估指标——即在推荐列表的不同深度（ k ）下向用户呈现正确解决方案的概率。针对配对位置问题，本系统在实际运行中会向用户提供6条建议，因此我们采用该阈值来定义系统的整体准确率。此外，我们定义 $NDCG^*$ 为首个正确建议的平均逆对数秩，以使用单一数值比较不同方法。这实际上是评估检索系统时常用的归一化折现累积增益的下界近似值，区别在于我们不会为超出首个正确预测后的后续正确预测额外计分。

6.1 Baselines

We evaluate our model against two types of baselines; pointcloud-based deep learning techniques that don't use CAD data, and simple ad-hoc heuristics.

Point Cloud Baselines. We chose to use point clouds as our comparison with discrete geometric representations because they are common in recent assembly modeling work [Huang et al. 2020; Li et al. 2020; Mo et al. 2019a, 2020b; Sung et al. 2017; Yang et al. 2020; Yin et al. 2020]. We evaluate against three popular methods; PointNet [Qi et al. 2017a], Pointnet++ [Qi et al. 2017b], and dynamic graph cnn (DGCNN) [Wang et al. 2019]. We structure these predictions following the siamese network design of ComplementMe [Sung et al. 2017], using a different output head for each task. Mate type learns an eight class classifier similar to our network, whereas mate location adopts the strategy of existing assembly modeling works and uses regression to predict an offset for each part's MCF; potential mates are then ranked by their MCFs' distance from these predicted offset, effectively snapping to the nearest potential mates. In order to give these algorithms the benefit of the user's face selection, we pre-rotate both parts into their final orientation and center and align the user's selection at the origin for the mate location task; for the mate type task, we transform the pair of parts into their final correct positions, re-centered with the mate at the origin aligned with the first part's mating coordinate frame. We expect that these methods will not do as well as ours because they do not have access to precise analytic geometry of the BREPs, and because they allocate their representation power across the geometry, whereas our method learns representations local to the topological entities on each part that will be interfacing. These models are also much larger than ours; they range from 834k parameters for PointNet to 1.5M for PointNet++, compared to only 126k for our model.

Ad-Hoc Baselines. Our ad-hoc baselines are task specific. For mate location we have three; *Random*, *Origin Type*, and *Snap to Selection*. *Random* is simply a random ranking of potential mates to get an expected performance floor. *Origin Type* ranks potential mates based on the pair of origin types for their MCFs, in order of the frequency we see such pairs in our dataset. *Snap to Selection* ranks mates by the distance to the center of mass of the user's selected faces; it is intended as a lower bound of the regress-and-snap approach without any regression. For mate type we have one ad-hoc baseline: *Label Distribution*. This always ranks mate type according to their frequency in our test set, ignoring the input.

6.2 Data Ambiguity

Location Ambiguities. Mate placement is a subjective task that depend on the user's design intent and the context they are working within. Our dataset only contains the mates that were needed to create the objects that the parts were used for, but parts can be used in other ways not seen in our dataset. In our quantitative metrics we evaluate strictly against the mates which appear in our dataset, so we asked a CAD expert to evaluate some of our system's prediction and indicate which suggestions were plausible uses of those parts. For all examples shown in the paper (Figures 13 and 18) we have

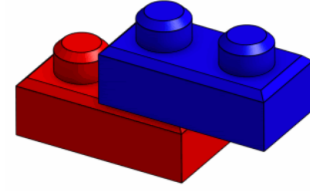


Fig. 11. Example of an ambiguous mate. The survey participants were divided between cylindrical (blocks can be rotated around the peg and slid off the peg), slider (blocks can be slid off the peg), and fastened (blocks should stay in place). In the dataset, this is classified as a parallel mate.

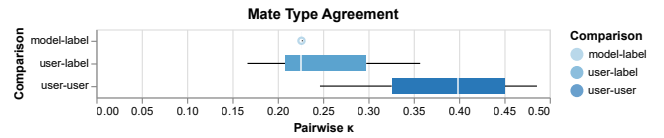


Fig. 12. Mate type agreement, measured by pairwise Cohen's κ , between surveyed CAD users, our model, and the dataset labels. Our model agrees with the labels to the same degree as the median expert, indicating that we achieve human performance. The experts agree within themselves to a greater degree than they agree with the dataset labels, indicating that there are additional contextual clues for mate type not captured by just looking at a pair of parts.

indicated with a green checkmark all mates that the CAD expert selected.

Mate Type Ambiguities. The mate type that a design uses is highly dependent on the designer's context. For example, our dataset over-represents *fastened* mates, but visual inspection finds gears marked as fastened rather than revolute; the user who created that assembly was not concerned with modeling the gear motion. To quantify the level of ambiguity on our dataset, we recruited seven expert CAD users to label 96 mates (12 of each mate type as labeled in our dataset) with what mate type they thought it should be. Overall, there was a majority consensus among the CAD users for 70 out of the 96 mates, which suggests we can expect to an upper bound of roughly 73% for top suggestion accuracy. Figure 11 shows an example of a mate that our experts gave three different labels to, and the dataset gave a fourth. We further quantified agreement using pairwise Cohen's κ , a 0-1 measure of agreement above random chance pictured in Figure 12. We measured agreement among the CAD experts, between the CAD experts and our dataset, and between our dataset and our model predictions. This analysis showed that while our CAD experts agree more with each other than with the dataset labels, our model's predictions agree with the dataset to the same degree as the median CAD expert, indicating that it achieves human-level performance.

6.3 Mate Location Prediction

Figure 13 shows a selection of location predictions from our system across a variety of complexities from a dozen possibilities to several thousand. Our model has learned to align key features like holes

6.1 基线

我们采用两种基准模型对我们的模型进行评估：一类是基于点云的深度学习技术，这些技术不使用CAD数据；另一类是简单的即兴启发式方法。

点云基线。我们选择使用点云作为与离散几何表示方法的对比基准，因为它们近期的装配建模研究中十分常见[Huang等人2020; Li等人2020; Mo等人2019a、2020b; Sung等人2017; Yang等人2020; Yin等人2020]。我们针对三种主流方法进行评估：PointNet[Qi等人2017a]、Pointnet++[Qi等人2017b]以及动态图卷积神经网络(DGCNN)[Wang等人2019]。这些预测结果采用ComplementMe[Sung等人2017]的孪生网络架构，并为每个任务设置独立的输出头。配对类型任务会学习一个类似于我们网络的八类分类器，而配对位置任务则采用现有装配建模工作的策略，通过回归方法预测每个部件的MCF偏移量；随后根据各潜在配对对象的MCF与这些预测偏移量之间的距离对其进行排序，从而有效锁定最近的潜在配对对象。为使这些算法能受益于用户的人脸选择过程，我们在配对位置任务中，先将两个部件预旋转至其最终朝向，并将用户的选定位置对齐至原点；对于配对类型任务，则将两部件组合转换至其最终正确位置，并以配对对象为原点，使其与第一部件的配对坐标系对齐。我们预计这些方法的效果可能不及我们的方法，原因在于它们无法获取BREP的精确解析几何信息，且其表征能力分散于整个几何结构之中，而我们的方法能够学习各部件上待对接拓扑实体的局部表征。此外，这些模型的规模也远大于我们的模型——例如PointNet模型的参数量可达834k，而我们的模型参数量仅为1。PointNet++的参数量为5M，而我们的模型仅为12.6万。

临时基准方法。我们的临时基准方法均针对特定任务设计。对于配对定位任务，我们设有三种方法：*随机*、*原始类型*以及*锁定于选区*。*随机*仅对潜在配对项进行随机排序以获得预期性能下限。*原始类型*根据配对项MCF中原始类型的组合频率，按其出现频次顺序对潜在配对项进行排序。*锁定于选区*则依据配对项与用户选定面部质心的距离进行排序；该方法旨在作为无需回归分析的“回归-锁定”方法的下限。对于配对类型，我们设有一个临时基准方法：*标签分布*。该方法始终根据测试集中配对类型的出现频率进行排序，忽略输入数据的影响。

6.2 数据歧义

位置歧义问题。配合件定位是一项主观任务，其结果取决于用户的设计意图及所处工作环境。我们的数据集仅包含为创建零件所用对象而所需的配合件，但零件也可能有其他未在数据集中体现的使用方式。在定量评估指标中，我们严格依据数据集中出现的配合件进行评估；为此，我们邀请了一位CAD专家对系统的部分预测结果进行评估，并指出哪些建议是这些零件

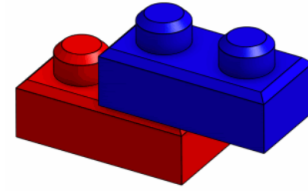


图11. 不明确配对示例。调查参与者被分为圆柱形（块可绕销钉旋转并从销钉上滑下）、滑动式（块可从销钉上滑下）和固定式（块应保持原位）三类。在数据集中，此类型被归类为平行配对。

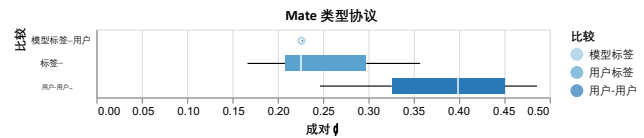


图12. 通过成对Cohen's κ 测量的调查CAD用户、我们的模型与数据集标签之间的配对类型一致性。我们的模型与标签的一致性程度与专家中位数一致，表明我们达到了人类水平的表现；专家们内部的一致性程度高于其与数据集标签的一致性，说明仅凭观察部件对无法捕捉到配对类型的额外上下文线索。

的合理使用方案。对于论文中展示的所有示例（图13和图18），我们已用绿色勾号标出了CAD专家所选择的配合件。

配合类型歧义问题。设计中采用的配合类型在很大程度上取决于设计师的具体应用场景。例如，我们的数据集过度偏向“固定配合”类型，但通过目视检查发现，某些齿轮被标注为“固定配合”而非“旋转配合”；创建该装配体的用户并未关注对齿轮运动的建模。为了量化我们数据集中的歧义程度，我们邀请了七位CAD专家对96个配合关系（即数据集中标注的每种配合类型各12个）进行标注，指出其应属于何种配合类型。总体而言，CAD专家们对于其中的70个配合关系达成了高度共识，这表明我们可以预期，顶级建议准确率的上线约为73%。图13展示了专家们对某配合关系给出了三种不同标注，而数据集则给出了第四个标注的示例。我们进一步采用Cohen's κ 指标来量化一致性——这是一种衡量实际一致性与随机概率之差的0-1型一致性度量指标，如图12所示。我们分别测量了CAD专家之间的一致性、CAD专家与数据集之间的一致性，以及数据集与模型预测结果之间的一致性。该分析表明，虽然我们的CAD专家之间的一致性高于与数据集标签的一致性，但我们的模型预测结果与数据集的吻合度与中位数CAD专家预测结果相当，表明其达到了人类水平的表现。

6.3 Mate定位预测

图13展示了我们系统在不同复杂度下（涵盖从十几种到数千种可能性）的若干位置预测结果。我们的模型已学会对诸如孔洞等关键特征进行对齐。

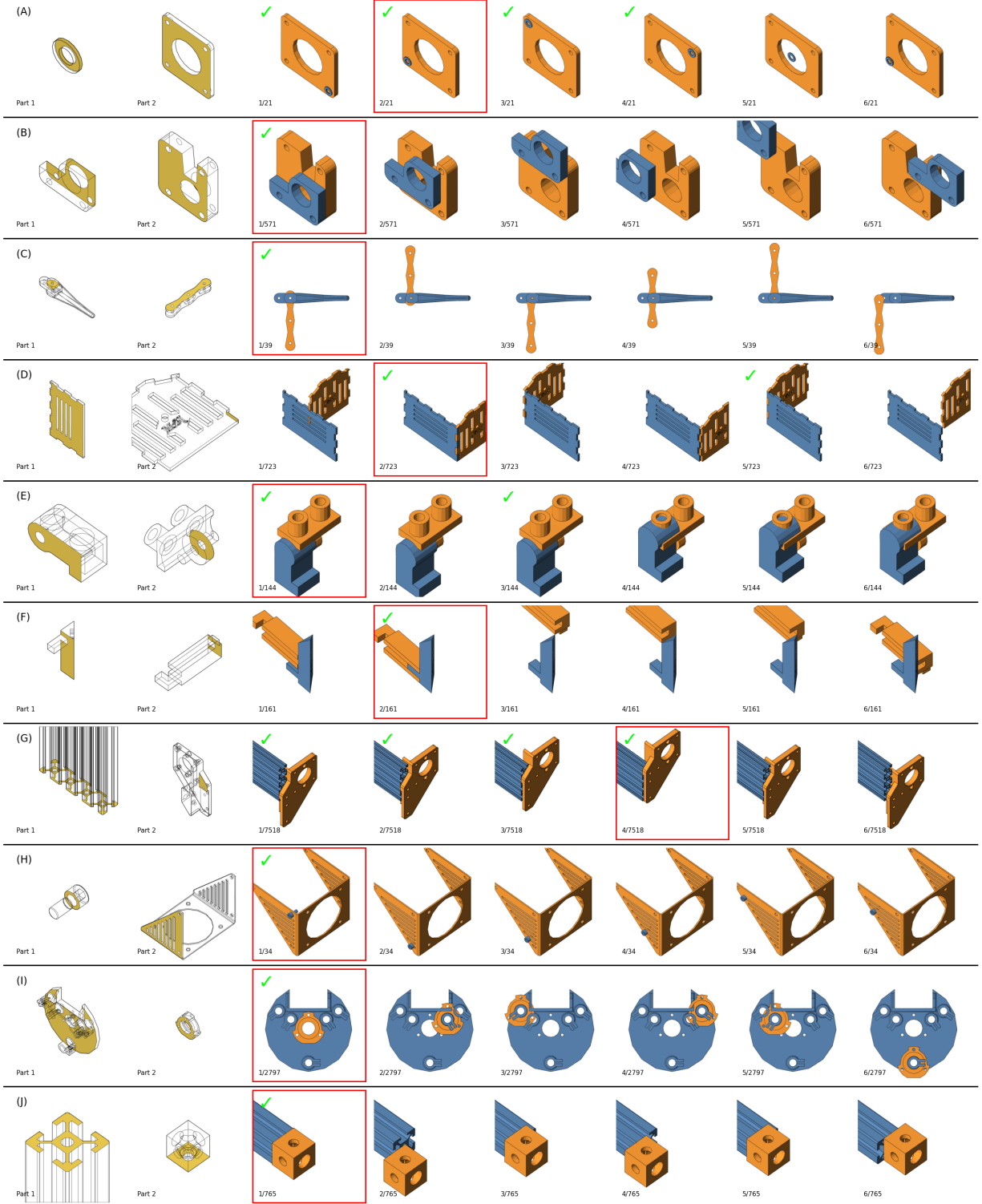


Fig. 13. Location predictions. Mates labeled as correct in our dataset are boxed, and mates that a CAD expert judged as good prediction have a checkmark. Prediction rank is indicated below. For these examples, we have excluded suggested mate locations that are equivalent to an earlier mate given the mate type to better reflect user preferences, which affects both rank and number of possibilities compared to our quantitative results.

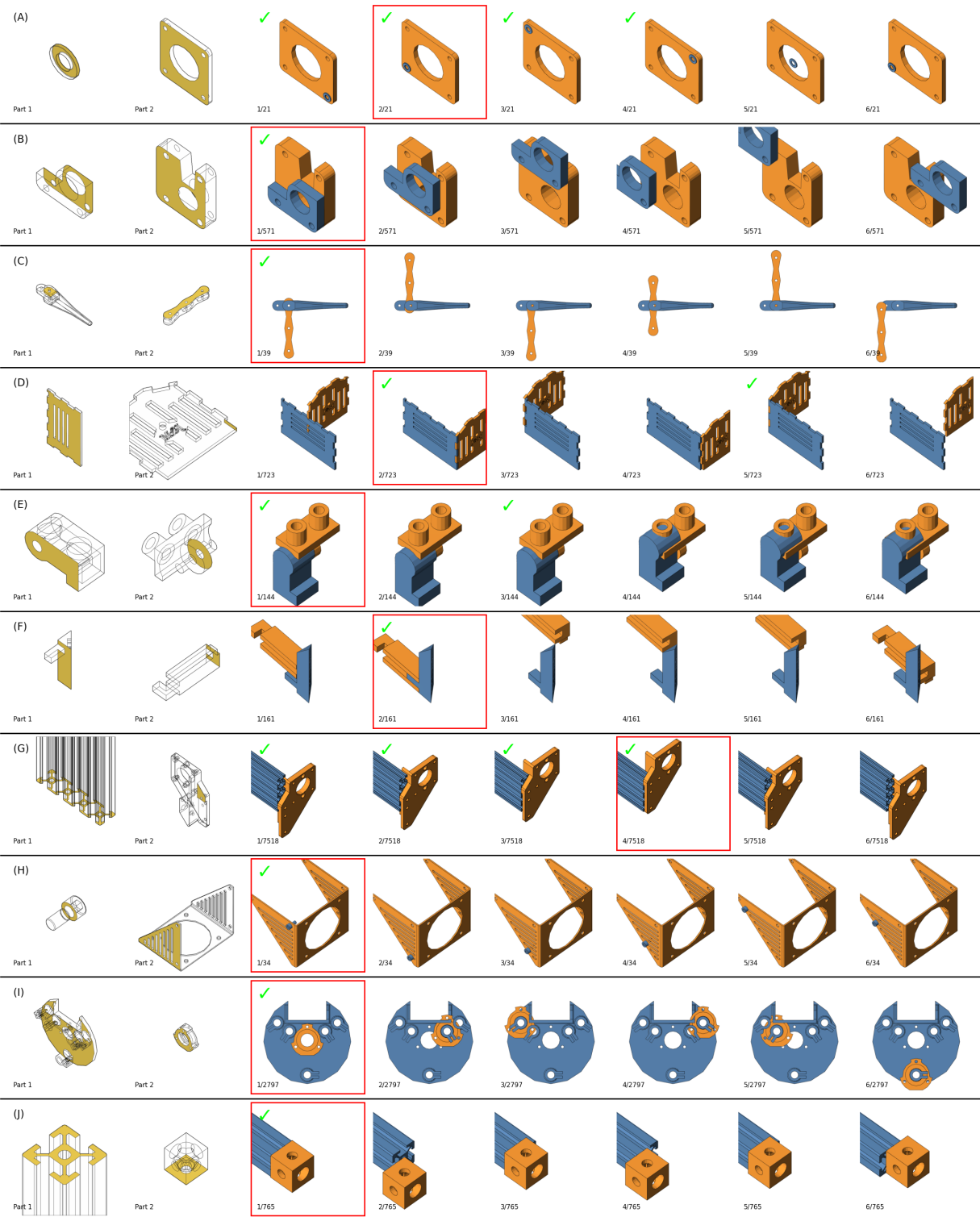


图13. 位点预测结果。在我们的数据集中被标记为“正确”的配对位置已用方框标出；而经CAD专家判定为“良好预测”的配对位置则标有勾号。预测排名信息列于下方。对于这些示例，我们已排除了根据配对类型与先前已有的配对位置等效的推荐配对位置，以更准确地反映用户偏好——这与我们的定量分析结果相比，会影响排名及可能的配对方案数量。

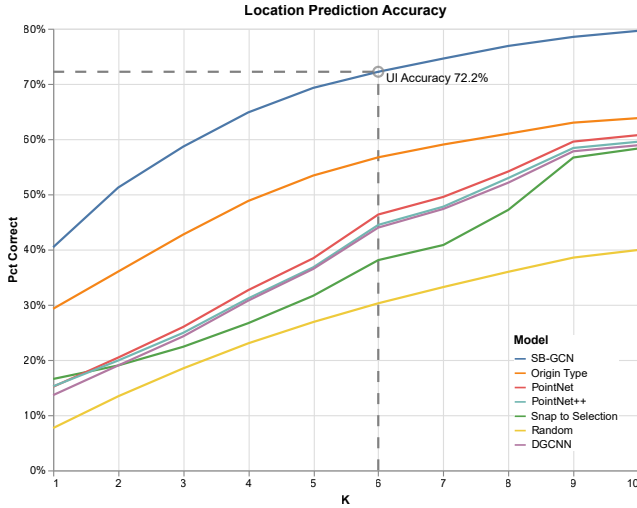


Fig. 14. Our model compared against baselines on mate location prediction task.

and edges of similar sizes. In example (H) we see the effect of not knowing the mate type in advance; the bolt holes and slots have a similar radius and our model suggests mates that could be either *revolute* (in the bolt holes) or *pin slots* (in the slots). Examples (E) and (I) show that our model learns to align features but does not learn to avoid part overlap.

Baseline Comparisons. Figure 14 illustrates the performance of our model in predicting mate location, as compared to several baselines. Our model outperforms all baselines, and achieves a 72.2% accuracy at our UI threshold of 6 predictions. The inference type baseline, our simplest model using CAD data, lags behind our system by roughly 15%, but still in turn outperforms all geometric baselines. While these geometric baselines do learn something; they perform better than a random selection baseline, they do not significantly outperform Snap to Selection, a model with no learning.

Scaling. We evaluate how well our method scales with the number of potential mates to rank. We binned prediction results by the number of potential mates (without the merging present in Figures 13 and 18, so these numbers are higher than appears in those figures), then computed quantile statistics of the rank of the first correct prediction, which we plot in Figure 15. The median rank grows much less quickly than the number of possible mates, and in fact stays below our interface cutoff of six suggestions for six of the eight buckets, showing that our predictor can scale well to more complex problems.

Failure Cases. Figure 18 illustrates some failures of our model. The far right column shows the first correct prediction and its rank. Examples (A)-(C) illustrate cases where additional context beyond the parts would be needed to correctly infer the mate location. In each of these cases, a third part interacts with the mated part. Our model did, however, pick alternatives that the expert CAD user validated as useful for some use case – just not the one seen in

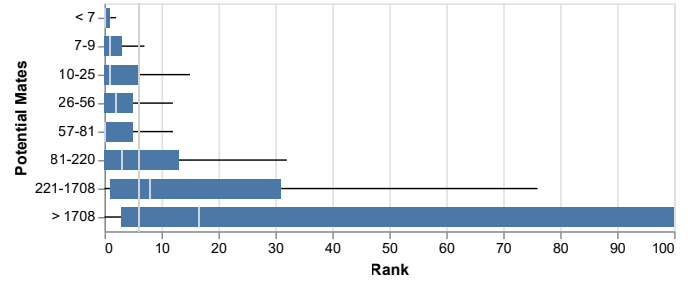


Fig. 15. Rank of the first correct mate suggestion as the number of potential mates grows. Ranges selected to have roughly even mate counts (3500), except the final bin which has only 646. While variance does grow with the number of potential mates, the median correct prediction rank grows much more slowly, and only exceeds our UI cutoff of 6 selections (light grey line) above 220 possibilities. Outliers have been excluded from whiskers, and the whisker for the final bar has been truncated.

our dataset. Example (D) shows a common problem when parts have multiple locations that need to be aligned, our model does not always align all of them; our chosen locations align one, two, or three holes, but miss the one location that would align all four. In example (E), the walls as modeled are actually hovering above the floor, and so are not topologically close enough to be captured in MCF representations. Both of these examples suggest we should incorporate more extrinsic geometric features and connectivity to account for mismatch between topological and geometric distances. In the final example, the horizontal board has no marks on its end, so our model attempts to match it with the midpoint and a similarly sized edge of the vertical one. The correct mate aligns the end of the horizontal board with a pre-drilled nail hole in the vertical board, and since it's a nail, there is no hole modeled in the endcap to suggest alignment.

6.4 Mate Type Prediction

Figure 16 illustrates performance against baselines for the mate type task. Our model outperforms all of the baselines, though the gap is not as significant as in the location task due to the tighter bounds between the label distribution baseline (presumed lower bound) and our ambiguity-based estimate of 73% first suggestion accuracy upper bound. Having access to definitive analytic information, for example that the MCFs are centered on a cylinder and a circle indicating likely rotational degrees of freedom, gives our model an edge over the pointcloud based methods that must infer this from samples. The 73% upper bound estimate is very rough due to a small sample size, so our model's 70% top suggestion accuracy and alignment with the median human expert (discussed above, see Figure 12) leads us to believe our model is approaching the limits of accuracy given the dataset ambiguity.

6.5 Ablations and Design Decisions

Analytic versus Discrete Geometry. Since we want to integrate with existing BREP based CAD software, and need the precision of analytic geometry to interface with CAM systems, any mate definitions we produce will need to be defined in reference to BREPs.

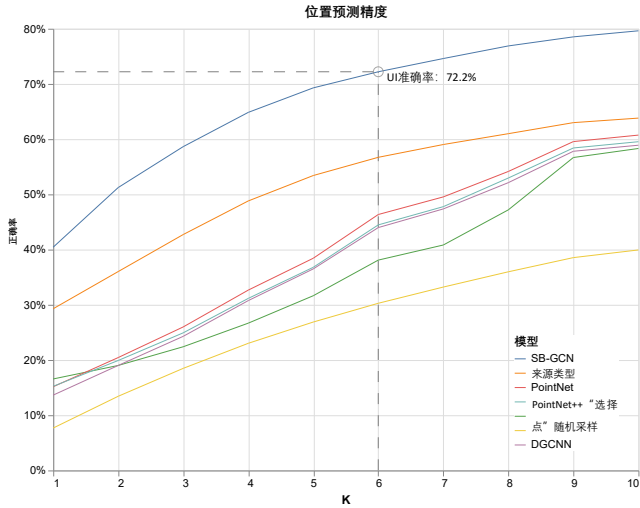


图14：我们的模型在配偶定位预测任务上与基线方法进行对比。

以及尺寸相似的边缘。在示例(H)中，我们观察到了事先未知配合类型所带来的影响：螺栓孔与槽孔具有相似的半径，我们的模型推测这些配合可能属于**旋转配合**（位于螺栓孔内）或**销槽配合**（位于槽口中）。示例(E)和(I)表明，我们的模型能够学习对特征进行对齐，但无法学会避免零件重叠。

基线对比。图14展示了我们的模型在预测配对位置方面的性能，并与若干基线方法进行了比较。我们的模型优于所有基线方法，在UI阈值为6次预测时，达到了72.2%的准确率。推理类型基线——即我们使用CAD数据的最简单模型——相较于我们的系统稍逊一筹，约落后15%，但仍优于所有几何基线方法。虽然这些几何基线方法确实有所学习；其性能优于随机选择基线，但并未显著超越“Snap to Selection”模型——该模型不具备任何学习能力。

可扩展性评估。我们评估了本方法在潜在配偶数量变化时的可扩展性表现。我们根据潜在配偶数量对预测结果进行分箱处理（未采用图13和图18中所示的合并方式，因此这些数值高于该图所示数值），随后计算首个正确预测的排名的分位数统计量，并将结果绘制于图15中。中位排名的增长速度远低于可能配偶数量的增长速度；实际上，在八个分箱中的六个分箱内，该值均低于我们的界面阈值（即六条建议），这表明我们的预测器能够良好地适应更复杂的场景。

故障案例。图18展示了我们模型的若干故障情况。最右侧列显示了首个正确的预测及其排名。示例(A)–(C)展示了需要借助部件之外的额外上下文信息才能正确推断配对位置的情形。在所有这些案例中，第三个部件均与已配对的部件存在交互作用。然而，我们的模型确实选择了专家CAD用户验证为某些用例中有效的替代方案——只不过并非我们在数据集中所见的情

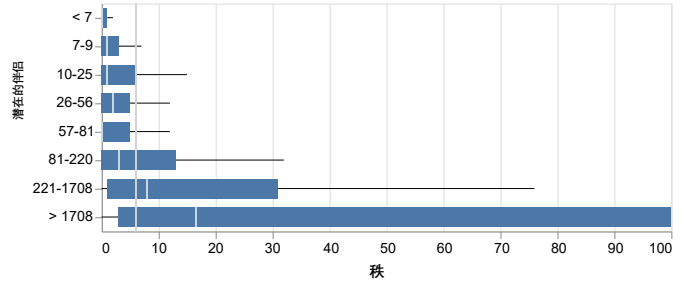


图15：随着潜在配偶数量的增加，首个正确配偶建议的排名变化情况。所选区间旨在使配偶数量大致均衡（3500），但最后一个区间仅包含646个选项。虽然方差会随潜在配偶数量的增加而增大，但中位正确预测排名的增长速度要慢得多——仅在选项数超过220时，该排名才会超过我们UI设定的6次选择阈值（浅灰色线）。异常值已从须状区段中剔除，且最后一个柱状图的须状区段已被截断。

况。示例(D)展示了当部件具有多个需对齐的位置时常见的问题：我们的模型并非总能对齐所有位置；所选位置仅能对齐一个、两个或三个孔位，却遗漏了可对齐全部四个孔位的那个位置。在示例(E)中，模型所构建的墙体实际上悬浮于地面之上，因此在拓扑结构上与地面的距离过远，无法被MCF表示所捕捉。这两个示例表明，我们应引入更多的外部几何特征和连通性信息，以校正拓扑距离与几何距离之间的不匹配。在最后一个示例中，水平板的末端没有标记，因此我们的模型尝试将其与垂直板的中点及一个尺寸相近的边缘进行匹配。正确的配对方式是将水平板的末端与垂直板上预钻的钉孔对齐；由于该部件为钉子，其端盖上并未设计任何可提示对齐的孔洞。

6.4 咖啡豆类型预测

图16展示了我们的模型在配对类型任务中的性能与基线方法的对比结果。尽管由于标签分布基线（假定下限）与我们基于模糊性的73%首次建议准确率上限估计值之间的差距较小，但我们的模型仍优于所有基线方法。能够获取明确的分析信息（例如MCF分布集中在圆柱体和圆上，这表明了可能的旋转自由度），使我们的模型相较于必须从样本中推断这些参数的点云方法更具优势。由于样本量较小，73%的上限估计值较为粗略；因此，我们模型达到的70%最高建议准确率以及人类专家中位数预测值的对齐程度（详见上文图12），使我们相信在当前数据集存在模糊性的情况下，我们的模型已接近准确性的极限。

6.5 消融术与设计决策

分析几何与离散几何。由于我们需要与现有的基于BREP的CAD软件实现集成，并且需要利用分析几何的精确性来与CAM系统对接，因此我们生成的所有零件定义均需参照BREP文件进行定义。

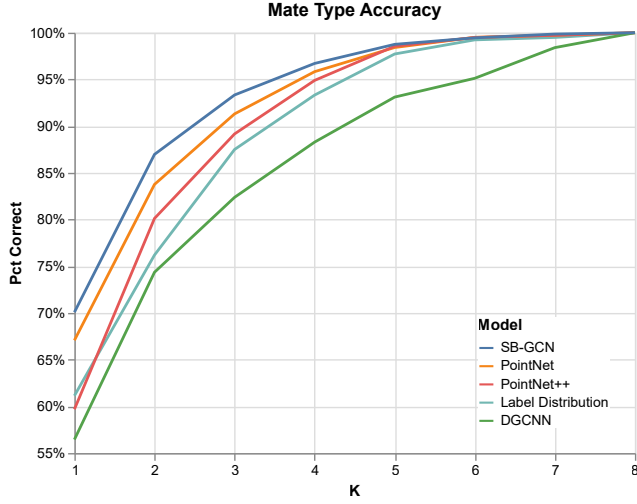


Fig. 16. Our model compared against baselines on mate type prediction task.

However, there is still a question of whether to use the BREP data directly for prediction, or to convert to a discrete geometric representation, like a point cloud or mesh, that is more commonly used in assembly modeling. In this alternative, we would predict relative offsets for each part, then project to the nearest BREP mate. Our baseline comparisons show that this approach does not work well for pointcloud based models that have previously been used in assembly modeling. To test the viability of the regress-and-snap strategy in general, we tried removing the learning component by testing against a noisy oracle that always predicts the true offsets, plus a controllable noise vector scaled to be a fraction of the standard deviation of offsets off all mates under consideration. Figure 17 shows that regress-and-snap is a losing strategy: accuracy drops at even very small amounts of output noise.

Feature and Network Selection. To determine which features to give our network, we incrementally added layers of features: just parametric functions, parametric function parameters, topological entity size, and bounding box plus center of mass and moment of inertia tensor. We also experimented with several convolutional architectures. The simplest was our *plain* network, which does not use the structural BREP data at all, instead learning embedding features with a shared MLP across topological entities. Our most direct analog to [Willis et al. 2020] is our *homogeneous* network. Similar to Willis et. al., we treat the BREP as an undirected graph and apply homogeneous message passing. However, since we need embeddings of all entity types, not just faces, we include all classes of entities in our graph. We also could not use grid-sampled face points as in Willis et. al. since non-face entities cannot be grid sampled. We tried adapting other sampling strategies to topological entities of lower dimensionality, but this strictly decreased performance and so was dropped. Since our BREP graphs are naturally heterogeneous, we also tried a heterogeneous variant of this network that learned

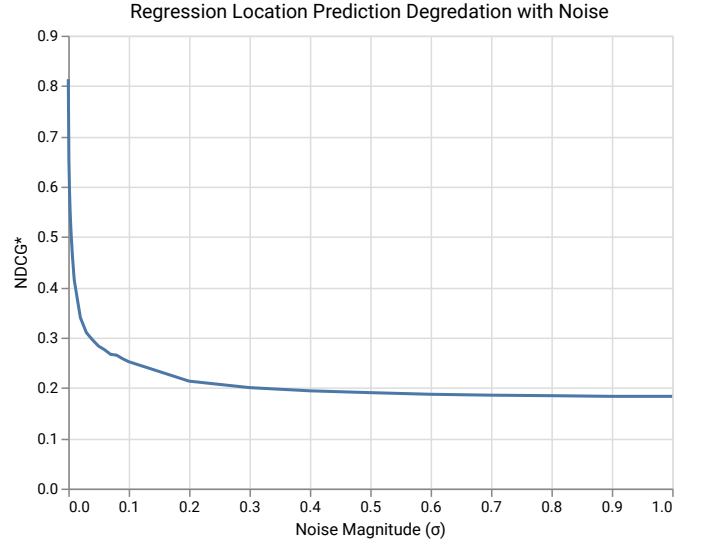


Fig. 17. Degradation of regression-based location accuracy in the presence of noise. The sharp accuracy drop-off in the presence of small prediction noise shows that regress-and-snap prediction cannot perform well for predicting mate coordinate frames.

Table 2. Validation NDCG* for mate type and location problem for experiments with or without a GCN, and with all input features or just the parametric function type.

	Mate Type		Location	
	Fn Type	All	Fn Type	All
No GCN	.832	.846	.470	.561
SB-GCN	.846	.850	.547	.576

a different message passing function for each type of edge (vertex-edge, edge-loop, loop-face, face-loop, loop-edge, and edge-vertex). This approach is more costly as it has six times the number of parameters. Finally, we tried the BREP structured network described in Section 5, which learns separate message passing functions for each edge type while having the same number of parameters as the homogeneous network, and fewer overall message passes than either homogeneous or heterogeneous. We found that these two axes of exploration, features and network convolution, both improve results for the location task independently, and that each can compensate for the other. Table 2 summarizes these differences. We hypothesize that the additional parameter and computed features capture some of the information that the network otherwise needs neighborhood data to infer. We choose to use our structured model with all features since it performs as well as other architectures that are more costly in computation and parameter count.

7 LIMITATIONS AND FUTURE WORK

In proposing the first learning model for suggesting CAD mates, our approach leaves several avenues of future work.

The inevitable existence of false negatives in our dataset means that our quantitative results under-represent the quality of our

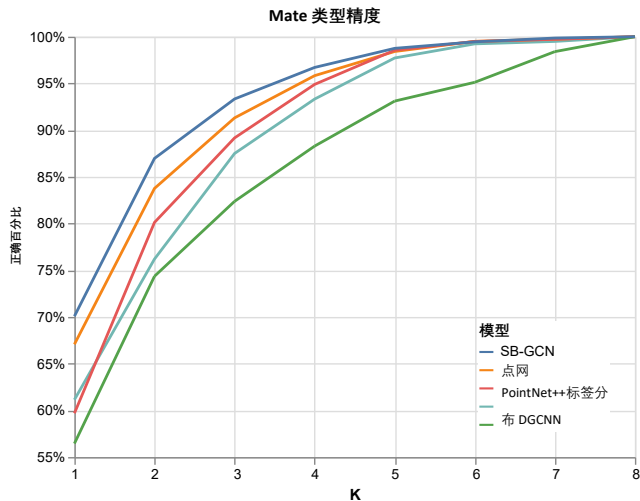


图16：我们的模型在配对类型预测任务上与基线方法进行对比。

然而，仍存在一个问题：是直接使用 BREP 数据进行预测，还是将其转换为离散几何表示（如点云或网格），这种表示方式在装配建模中更为常用。在第二种方案下，我们先为每个零件预测相对偏移量，再将其投影至最近的 BREP 配对对象。我们的基线比较表明，该方法对于此前已应用于装配建模的基于点云的模型并不适用。为了全面检验“回归-snapping”策略的可行性，我们尝试通过使用一个含噪声的“真知”来测试——该“真知”始终能预测真实的偏移量，并附加一个可控制的噪声向量，其幅度设定为所有待处理配对对象偏移量标准差的某个比例。图17显示，“回归-snapping”策略是一种失败的策略：即使输出噪声非常小，其准确率也会下降。

特征与网络选择。为确定应为网络添加哪些特征，我们逐步增加了多层特征：仅参数化函数、参数化函数参数、拓扑实体尺寸、边界框以及质心和转动惯量张量。我们还尝试了多种卷积架构，其中最简单的是我们的纯网络——该网络完全不使用结构 BREP 数据，而是通过跨拓扑实体共享的 MLP 学习嵌入特征；与[Willis等人2020]研究中最直接的对应模型是我们的同质网络。与Willis等人类似，我们将 BREP 视为无向图并应用同质消息传递机制。但由于我们需要所有类型实体（而不仅限于面部）的嵌入表示，因此在图中包含了所有实体类别。此外，我们无法像Willis等人那样使用网格采样面部点，因为非面部实体无法进行网格采样。我们曾尝试针对低维度拓扑实体调整其他采样策略，但这会显著降低性能，故最终放弃该方案。由于我们的 BREP 图本身具有天然的异构性，我们还尝试了该网络

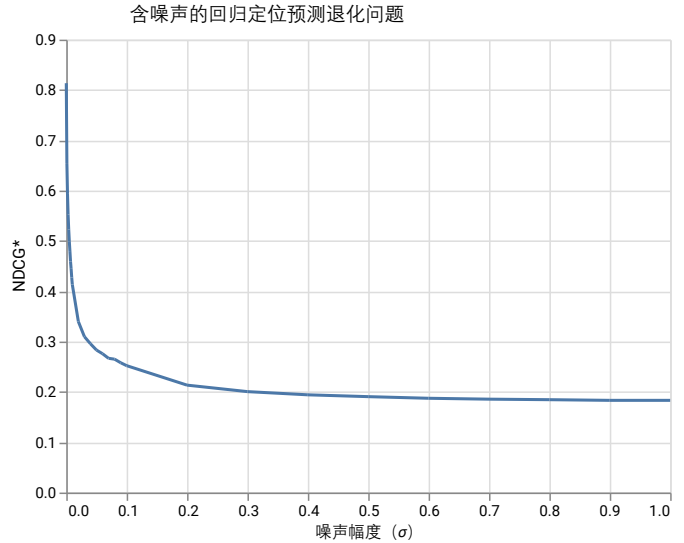


图17. 在存在噪声的情况下，基于回归的位置精度下降情况。在存在微小预测噪声时，精度的急剧下降表明，“回归-定点”预测方法无法有效预测配对坐标系。

表2.针对包含或不包含 GCN、以及包含全部输入特征或仅包含参数函数类型的实验，配偶类型与位置问题的验证 NDCG *。

	茶类		地点	
	Fn类型	所有	Fn类型	所有
无 GCN	.832	.846	.470	.561
SB-GCN	.846	.850	.547	.576

的一种异构变体，为每种边类型（顶点-边、边-环、环-面、面-环、环-边以及边-顶点）学习不同的消息传递函数。这种方法成本更高，因为其参数数量是同构网络的六倍。最后，我们采用了第5节所述的 BREP 结构化网络：该网络为每种边类型学习独立的消息传递函数，同时参数数量与同构网络相同，且整体消息传递次数少于同构或异构网络。我们发现，这两个探索方向——特征提取和网络卷积——都能独立提升定位任务的性能，且彼此可相互弥补。表2总结了这些差异。我们假设额外的参数和计算特征能够捕捉网络原本需要依赖邻域数据才能推断的部分信息。我们选择使用包含所有特征的结构化模型，因为其性能与其他在计算复杂度和参数数量上更为高昂的架构相当。

7 局限性与未来工作

在提出首个用于推荐CAD配对部件的学习模型时，我们的方法为未来的研究指明了若干方向。

我们数据集中不可避免地存在假阴性结果，这意味着我们的定量分析结果未能充分反映实际质量水平。

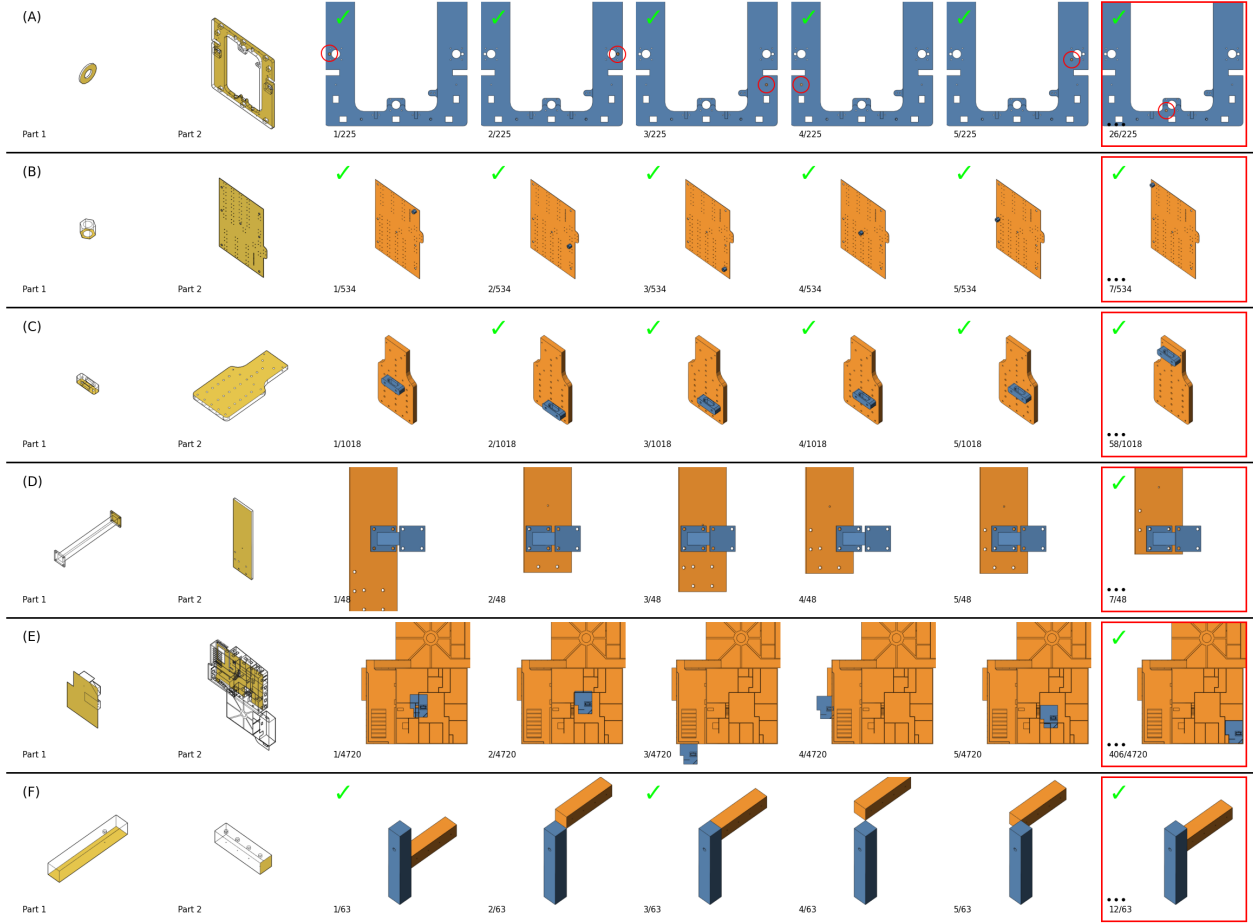


Fig. 18. Various failure cases. (A)-(C) highlight cases where our model produces plausible results, but does not have the greater context to understand the user's intent. (D) and (E) are cases where a better understanding of extrinsic geometry would help: in (D) to understand multiple-hole alignment, and in (E) to capture geometrically close features (the walls), which are topologically distant from the pertinent MFC reference (the floor). (F) shows a case where the material context would help; the are square lumber, and the original creator intended to align one piece with a pre-drilled nail hole in the other.

predictions, as evidenced by the expert annotations in Figure 18. While no *in the wild* dataset can ever capture all ways that a user may wish to use a given part pair, data augmentation can be used to reduce false negatives. In this work we augmented our dataset at the MCF level by computing alternative constructions for the same coordinate frames. Future work could extend this in three ways. First, as shown in Figure 19 (a), merging mates with different MCF locations that combine to the same relative positions if the mate type is given gives us a more accurate picture of result quality (as done in Figures 13 and 18). While this cannot be done as a post process if mate type is not known at inference (as reported in Figure 14), additional work could be done to augment the database finding all pairs with all equivalent MCFs. Second, as shown in Figure 19 (b), our work does not capture the alternative uses of pairs of parts that exist in our database because we separate our training and validation examples by mate pair rather than part pair. Unifying these and separating our examples by part pair would require extending our fingerprint for identifying equivalent BREPs to a full BREP matching

including a mapping between BREP topological entities. Finally, there are symmetries that we could exploit to augment our data with additional plausible mates; in Figure 10 this would result in the center MCFs of each hole being marked as ground truth since that part has 4-fold rotational symmetry.

Several additional features are required to make our system viable in a CAD workflow. Additional user interactions to select non-face topologies would remove the restriction of only selecting face orientations. Some mate suggestions are clearly non-viable due to intersections or displacements, these could be heuristically pruned to improve suggestion quality. We would also train an additional categorical predictor for mate location to allow for non-canonical part orientations, which together with non-face selection would allow our system to work with the vast majority of mates.

We would also like to include some discrete geometry features into our model. While point density around sharp features presents a challenge for PointNet features, other models such as MeshCNN [Hanocka et al. 2019] or the curve and surface convolution features

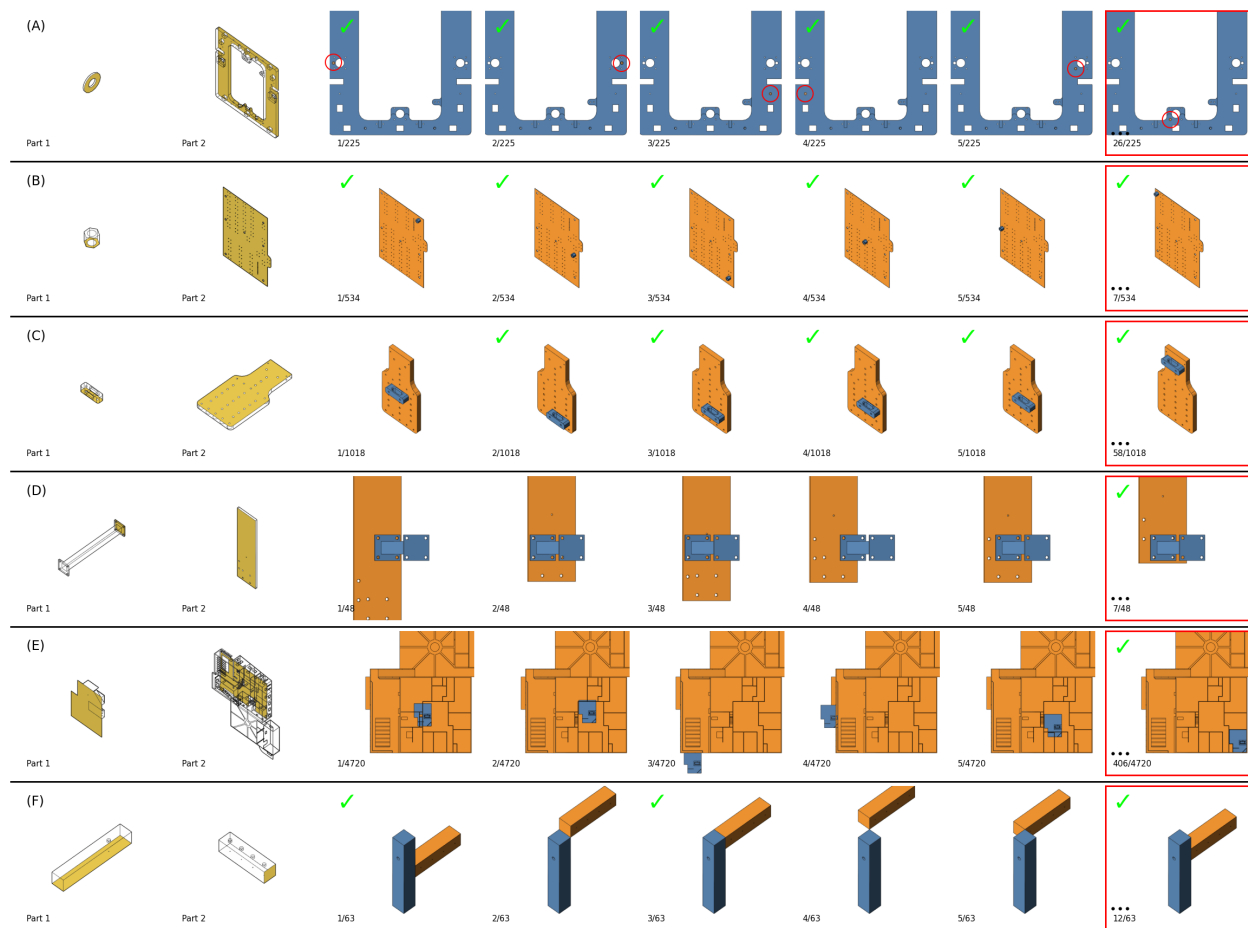


图18展示了多种故障案例。(A)至(C)突显了我们的模型虽能得出合理结果，但缺乏理解用户意图所需的整体背景信息的情况；(D)和(E)则表明更深入地理解外部几何结构将有助于解决这些问题：在(D)中需理解多孔洞的对齐问题，在(E)中需捕捉几何形状相近但拓扑位置远离相关MFC参考面（地板）的特征（即墙体）；(F)则展示了一个需要材料背景信息辅助的案例：所涉及材料为方形木材，原始设计者意图使其中一块木材与另一块木材上的预钻孔对齐。

预测结果如图18中的专家标注所示。虽然野外数据集无法涵盖用户可能希望使用给定零件对的所有方式，但数据增强技术可用于减少假阴性结果。在本研究中，我们在MCF层面对数据集进行了增强处理——即为相同的坐标系计算替代构造。未来工作可从三个方面扩展此方法：首先，如图19(a)所示，若已知配对类型，则将具有不同MCF位置但能组合成相同相对位置的配对进行合并，可更准确地评估结果质量（如图13和18所示）；然而，若在推理阶段未获知配对类型，则无法通过后处理实现此操作（如图14所示），此时可开展额外工作以增强数据库，找出所有具有相同等效MCF的配对；其次，如图19(b)所示，本研究未能捕捉数据库中存在的零件对的其他用法，这是因为我们是按配对类型而非零件对来划分训练与验证样本的。要统一这些方法并根据零件对分离我们的示例，就需要扩展我们的指纹识别等效BREPs的方法，以实现完整的BREP匹配，包括

BREP 拓扑实体之间的映射。最后，我们还可以利用对称性来增强数据，添加额外的合理配对；如图10所示，这将导致每个孔的中心MCF被标记为真实值，因为该零件具有四重旋转对称性。

要使我们的系统能够在CAD工作流程中有效运行，还需要具备若干附加功能。新增用户交互选项以选择非面拓扑结构，可突破仅限于选择面朝向的限制；部分配合建议因存在交线或位移问题而明显不可行，可通过启发式算法进行筛选以提升建议质量；我们还将训练一个专门用于预测配合位置的分类模型，以支持非标准零件朝向——结合非面选择功能后，该系统将能够兼容绝大多数类型的配合关系。

我们还希望将一些离散几何特征纳入我们的模型中。虽然尖锐特征周围的点密度对PointNet特征提取构成了挑战，但其他模型如MeshCNN [Hanocka等2019]或UV-Net的曲线与曲面卷积特征[Jayaraman等2021]则能

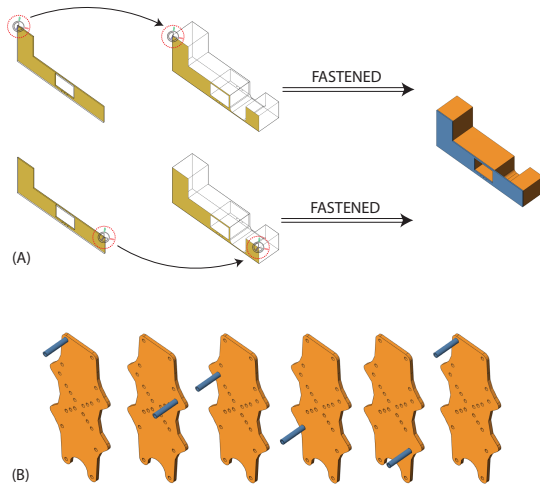


Fig. 19. Two types of false negatives present in our training data. (a) The same relative degrees of freedom for a part pair can be achieved by different combinations of MCFs, dependent on the mate type; we currently only capture those with equivalent MCFs. (b) Identical parts pairs appear in multiple configurations in our dataset, but since we train and validate our model *per-mate*, these alternatives appear as false negatives. If we can match topological entities across equivalent BREPs, we could unify these examples.

of UV-Net [Jayaraman et al. 2021]. Further, incorporating the greater assembly context into our predictions is an exciting challenge that we leave to future work (see Figure 18 (A)-(C)).

Our system is built to work with the Parasolid modeling kernel and with MCF based mates, but CAD systems use a variety of modeling kernels and mating schemes. Adapting our approach to other modeling kernels is straightforward; the primitives for which we support parametric features and MCF references are common to every BREP modeling kernel, and other entity types are only featurized by a categorical variable and geometric summary information. Adapting our data set to other modelers may incur data loss since different primitives may be used to generate the same geometry, leading to different potential MCFs. We will initially release our dataset and code in both Parasolid format and a pre-featurized serialization to avoid dependence on a modeling kernel, and plan to also make it available in STEP format with an OpenCascade implementation of BREP featurization.

Finally, this work lays the machine learning foundation for automatic mating interfaces and leaves open exciting avenues for expanding the capabilities of these systems. We limited our mate location search in this work to the neighborhood of the user's selection both to incorporate the user's design intent, and to limit the number of mates under consideration. If we expand the number of MCFs under consideration in each part, the number of possibilities grows quadratically, and will exhaust available memory at training time. This could be somewhat alleviated by random negative sampling, but even in the one-face neighborhood we had to cap each example to at most 10k MCF pairs to limit memory consumption, and some face selections can result in millions of pairs. Future work

can address this limitation by making the dependency linear, choosing a constant number of MCFs on one selected part as the mate location intent, then considering all MCFs of the other part. Another direction is to extend the system to include part selection—given an MCF of a single part, suggest other parts that could mate at that location. A system combining part retrieval with placement has the potential to drastically speed up the CAD mating workflow.

8 CONCLUSION

In this work we proposed SB-GCN the first heterogeneous graph network for BREPs capable of learning embeddings for many types of topological entities. We used SB-GCN to build the first assembly modeling tool aligned to the unique needs of the CAD workflow; it takes BREPs as input, is category agnostic, models assemblies as a collection of constraints, and defines those constraints in reference to analytic geometry. While we applied our network to problems of mating, and to one particular model of mates, we designed our model to learn to represent the core building block of all BREP-based CAD software: topological entities. Because of this, SB-GCN should be applicable to other mating systems, and other CAD tasks. Finally, we collected and cleaned the first BREP-based assembly modeling dataset, which we are releasing in conjunction with our model and baseline tasks. We are excited to see what new innovations in CAD will be built on this foundation.

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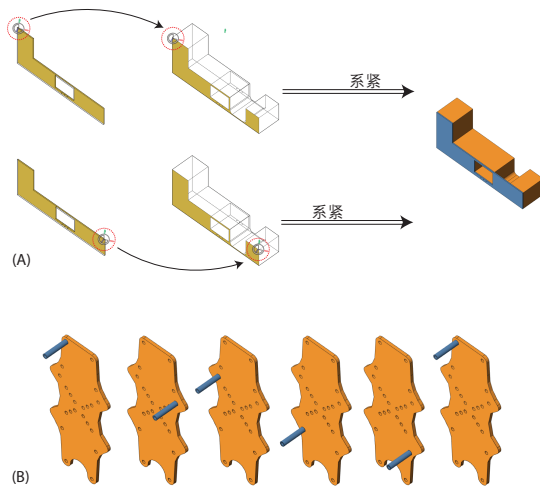


图19. 我们的训练数据中存在两种类型的假阴性结果。(a) 同一远端配对可由不同MCF组合实现，具体取决于配对类型；目前我们仅捕获那些具有等效MCF的配对。(b) 数据集中会出现多个配置下的相同部件对，但由于我们按每种配对对模型进行训练和验证，这些替代方案会表现为假阴性。若能跨等效BREPs匹配拓扑实体，则可统一处理此类示例。

有效应对这一问题。此外，将更广泛的装配上下文融入我们的预测过程是一个令人振奋的挑战，我们将此课题留待未来研究（参见图18(A)-(C)）。

我们的系统专为与Parasolid建模内核及基于MCF的配对机制兼容而设计，但CAD系统通常采用多种不同的建模内核和配对方案。将我们的方法适配至其他建模内核十分直接：我们支持参数化特征和MCF参考的几何基元是所有BREP建模内核共有的；其他实体类型仅通过分类变量和几何摘要信息进行特征化。若将数据集适配至其他建模软件，可能会导致数据丢失——因为生成相同几何体时可能使用不同的基元，从而产生不同的潜在MCF配置。我们将首先以Parasolid格式及预特征化的序列化格式发布数据集和代码（以避免依赖特定建模内核），并计划同步提供STEP格式版本，并集成OpenCascade实现的BREP特征化功能。

最后，本研究为自动配对接口奠定了机器学习基础，并为扩展这些系统的功能开辟了令人振奋的途径。在本研究中，我们将配对位置搜索范围限定于用户选择区域的邻域内——既是为了体现用户的设计意图，也是为了限制待考虑的配对数量。若我们在每个部件中扩大待考虑的MCF数量，可能使可能性数量呈二次方增长，从而耗尽训练时可用的内存空间。这一问题可通过随机负采样得到一定程度缓解，但即便在单面邻域范围内，我们也必须将每个示例的配对对数上限设为最多10,000对，以控制内存消耗；而某些面选择可能会产生数百万对配对组合。未来的研究可通过使依赖关系线性化来解决此局限性：

即在选定的某个部件上，选取一个固定数量的MCF作为配对位置意图，进而考虑另一部件中的所有MCF。另一研究方向是扩展系统功能以支持部件选择——当给定单个部件的MCF时，可建议其他能在该位置进行配对的部件。结合部件检索与定位功能的系统有望显著加速CAD配合设计流程。

8 结论

在本研究中，我们提出了SB-GCN——首个适用于BREPs的异构图网络，能够为多种类型的拓扑实体学习嵌入表示。我们利用SB-GCN构建了首款符合CAD工作流独特需求的装配建模工具：该工具以BREPs作为输入，具有类别无关性，并可对装配模型进行建模。约束集是一组约束的集合，用于定义这些约束。本文涉及解析几何领域。虽然我们将网络应用于装配问题及某一特定装配模型，但我们设计该模型旨在学习表征所有BREP基CAD软件的核心构建模块——拓扑实体。基于此，SB-GCN应可适用于其他装配系统及各类CAD任务。最后，我们收集并清理了首个BREP基装配建模数据集，并将该数据集与我们的模型及基准任务一并发布。我们满怀期待，见证基于这一基础所催生的CAD领域新创新成果。

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A DATA MODEL

Node Types. Table 3 summarizes the parametric functions in our system, along with their parameters. We exclude the origin parameter because these do not need to correspond to the topological entity location once boundaries are taken into account, and in practice are often far from surfaces, resulting in noise.

Origin Types. Each MFC has an *origin type* that specifies how the origin is computed from its reference topological entity. Table 4 lists all of the available origin types, along with the types of geometry they are applicable. Geometry without supported origin types are not able to be referenced to define MFCs. Note that all of these computations except *center* and *point* rely on boundary information, and so are not computable from the parametric geometry a topological entity in isolation, highlighting the need to integrate boundary information to properly understand the mating coordinate frames. Also note that the origin type is highly correlated with the geometry type of the referenced topological entity, which is why it is a useful proxy for our Origin Type baseline.

Computing MCF Orientation. The orientation of an MCF is defined by reference to a supported topological entity. Table 5 lists the geometric functions from which orientations can be determined, and describes how they are computed.

The orientation reference of an MCF only determines the frame's z-axis. The y-axis is computed as the cross product of this z-axis with the x-axis of the coordinate from that the part was modeled in, or the cross product with its y-axis if that x-axis is parallel to the chosen z direction.

B ABLATIONS

We tried sixteen combinations of graph architecture and feature set across the mate location and mate type problem. Figure 20 shows all of these results, compared along either axis. The mate type problem does not show much difference between methods, but for mate location there is a clear trend that having a GCN is better than not, that more features are better, and that adding features can partly make up for not using a GCN.

A 数据模型

节点类型。表3汇总了我们系统中的参数化函数及其对应参数。我们排除了原点参数，因为一旦考虑边界因素后，这些参数无需与拓扑实体的位置相对应；在实际应用中，它们通常距离曲面较远，从而产生噪声。

原点类型。每个MFC都具有一个*原点类型*，该类型用于指定如何根据其参考拓扑实体计算原点。表4列出了所有可用的原点类型及其适用的几何类型。不具备支持的原点类型的几何对象无法被引用以定义MFCs。需注意，除*中心*和*点*外，所有上述计算均依赖于边界信息，因此无法仅从孤立的参数化几何体中计算得出——这凸显了整合边界信息以正确理解装配坐标系的必要性。还需注意，原点类型与所引用拓扑实体的几何类型高度相关，这也是为何将其作为“原点类型”基准的有效代理指标。

计算MCF方向。MCF的方向是参照所支持的拓扑实体来定义的。表5列出了可用于确定方向的几何函数，并说明了其计算方法。

MCF的定向参考仅用于确定框架的z轴。y轴可通过以下方式计算：将该z轴与部件建模所用坐标系的x轴进行叉积；若该x轴与所选z方向平行，则可采用该x轴与y轴的叉积来计算y轴。

B ABLATIONS

我们针对交配位点与交配类型问题，尝试了图结构与特征集的十六种组合方案。图20展示了所有这些结果，并沿各维度进行了对比分析。在交配类型问题上，不同方法之间差异不大；但在交配位点问题上则呈现出明显趋势：使用 GCN 优于不使用；特征数量越多越好；增加特征可部分弥补未使用 GCN 带来的不足。

Table 3. Parametric functions and their parameters.

Function	Topology Type	Parameters	Description
plane	face	origin , normal	A plane.
cylinder	face	origin , axis, radius	A cylinder.
cone	face	origin , axis, half-angle	A cone.
torus	face	origin , axis, major-radius, minor-radius	A torus.
bsurf	face		A NURBS surface.
swept	face		Surface swept along a curve.
spun	face		Surface of rotation about an axis.
blend	face		Surface blending two other surfaces.
outer loop	loop		The outer trimming curve of a face.
inner loop	loop		An inner trimming curve of a face.
line	edge	origin , direction	A line.
circle	edge	origin , normal, radius	A circle.
ellipse	edge	origin , normal, major axis direction, major axis, minor axis.	An ellipse.
bcurve	edge		A B-spline curve.
icurve	edge		The intersection of two surfaces.
spcurve	edge		Curve embedding in a surface.
tcurve	edge		Trimmed curve.
pline	edge		A polyline.
point	vertex	origin	A point.

Table 4. Supported origin types for mating coordinate frames.

Origin Type	Supported Functions	Description
center	circle, ellipse, loop	The center of a curve or inner loop.
centroid	plane	The centroid of a planar face.
mid point	line	The mid point of a line segment.
point	point, cone	The location of a point or vertex of a cone.
top axis point	cylinder, torus, cone, spun	The most positive point of a surface of revolution projected onto its axis.
mid axis point	cylinder, torus, cone, spun	The mid-point of a surface of revolution projected onto its axis.
bottom axis point	cylinder, torus, cone, spun	The most negative point of a surface of revolution projected onto its axis.

Table 5. Z-axis computation for supported topological entities.

Topological Entity Function(s)	z-Axis
plane	normal
cylinder, cone, torus, spun	axis parameter
inner loop	plane normal if inner loop of a plane
line	along edge in positive parameterization direction
circle, ellipse	plane normal if adjacent to a plane, otherwise axis parameter
point	z-axis of CS part was modeled in.

表3. 参数函数及其参数。

功能面圆	拓扑类型：面	参数	描述
柱锥体	→面→面→面	ori正常含量 origin、轴、半径 origin、轴、半角 origin、轴、主半径、次半径	一架飞机。 圆柱体。 一个圆锥体。 环面。 一个NURBS曲面。
环面	→面→面→面		沿曲线扫描的曲面。绕轴旋转形成的曲面。
BSurf	→面→面→面		将两个表面进行融合。面的外轮廓曲线。面的内轮廓曲线。一条直线。
扫掠式纺	→面→环→环		一个圆。
纱混纺外	→边→边→边	ori金，方向；指引 ori金，正常，半径 ori长轴方向、长轴、短轴。	椭圆。
环、内	→边→边→边		一条B样条曲线。
环、线	→边→边→边		两个曲面的交线；曲线在曲面上的嵌入；修剪曲线。
圈、圆	→顶点		多段线。
环、椭圆、B型曲线、I型曲线			一个要点。
spcurve			
tcurve			
pline			
观点			

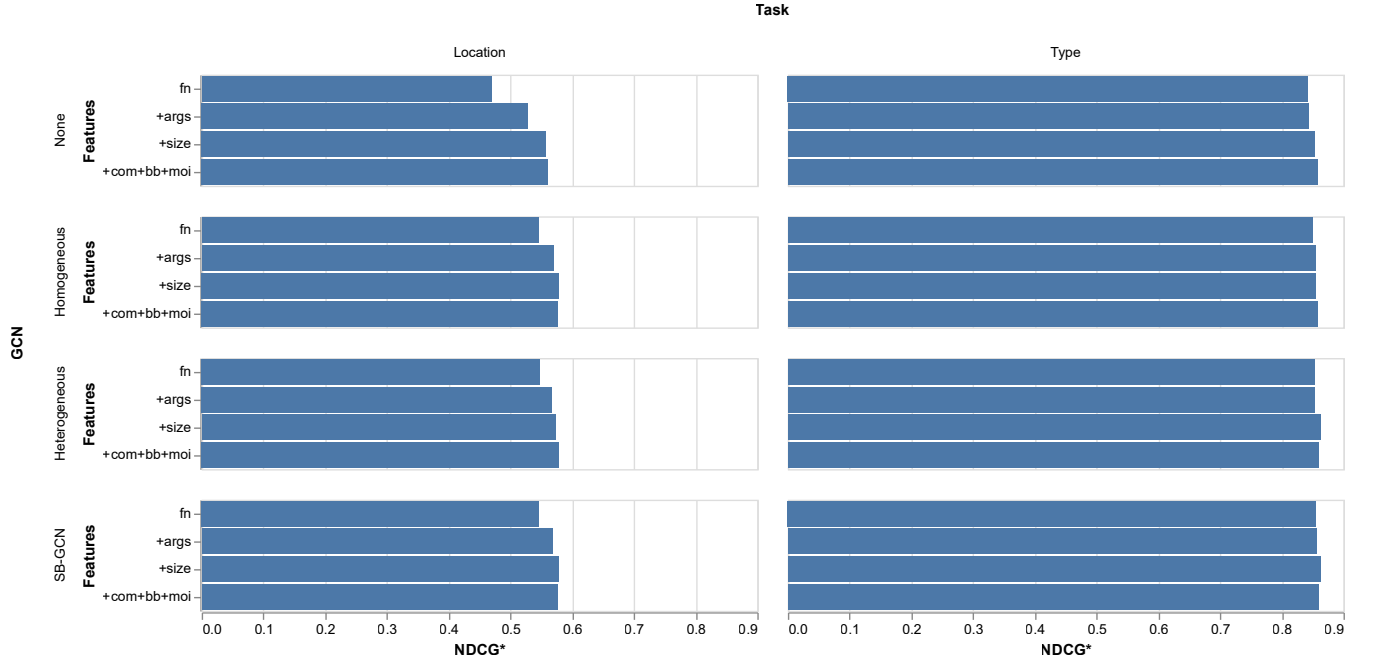
表4. 用于配对坐标系的支持原点类型。

来源类型	支持的功能	描述
中心；核心	圆、椭圆、环	曲线或内环的中心点。
重心	水平	平面面的质心。
中点	线条	线段的中点。
观点	点，圆锥	圆锥顶点或顶点的位置。
顶部轴点	圆柱体、环面、圆锥体；经过旋转处理的几何形状。	旋转曲面投影至其轴线上的最正面。
中轴点	圆柱体、环面、圆锥体；经过旋转处理的几何形状。	旋转曲面投影到其轴上的中点。
底部轴点	圆柱体、环面、圆锥体；经过旋转处理的几何形状。	旋转曲面投影到其轴上的最不利特征点。

表5. 支持的拓扑实体的Z轴计算。

拓扑实体函数（多个）	z轴
水平	正常的
圆柱体、圆锥体、环面、旋转内环线圆、椭圆点	轴参数为平面内沿边缘方向的内环时，其法线指向正参数化方向；若与平面相邻则法线指向该平面；否则轴参数为CS部分的z轴时，该轴被纳入建模。

(a) Feature Comparison



(b) GCN Comparison

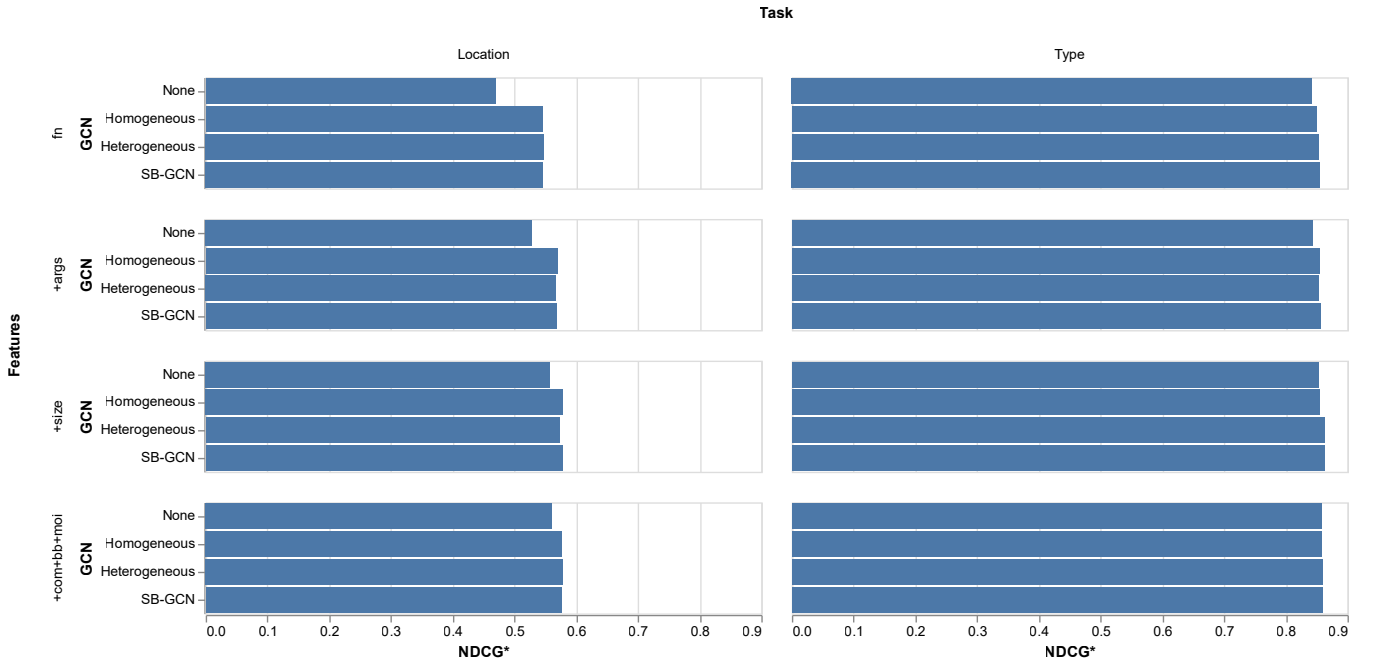


Fig. 20. Validation NDCG* for GCN type and feature set experiments trained against 70k subset. NDCG* is a modified NDCG score that stops counting at the first correct mate. For mate type this is equivalent to NDCG since there is only one correct mate type.

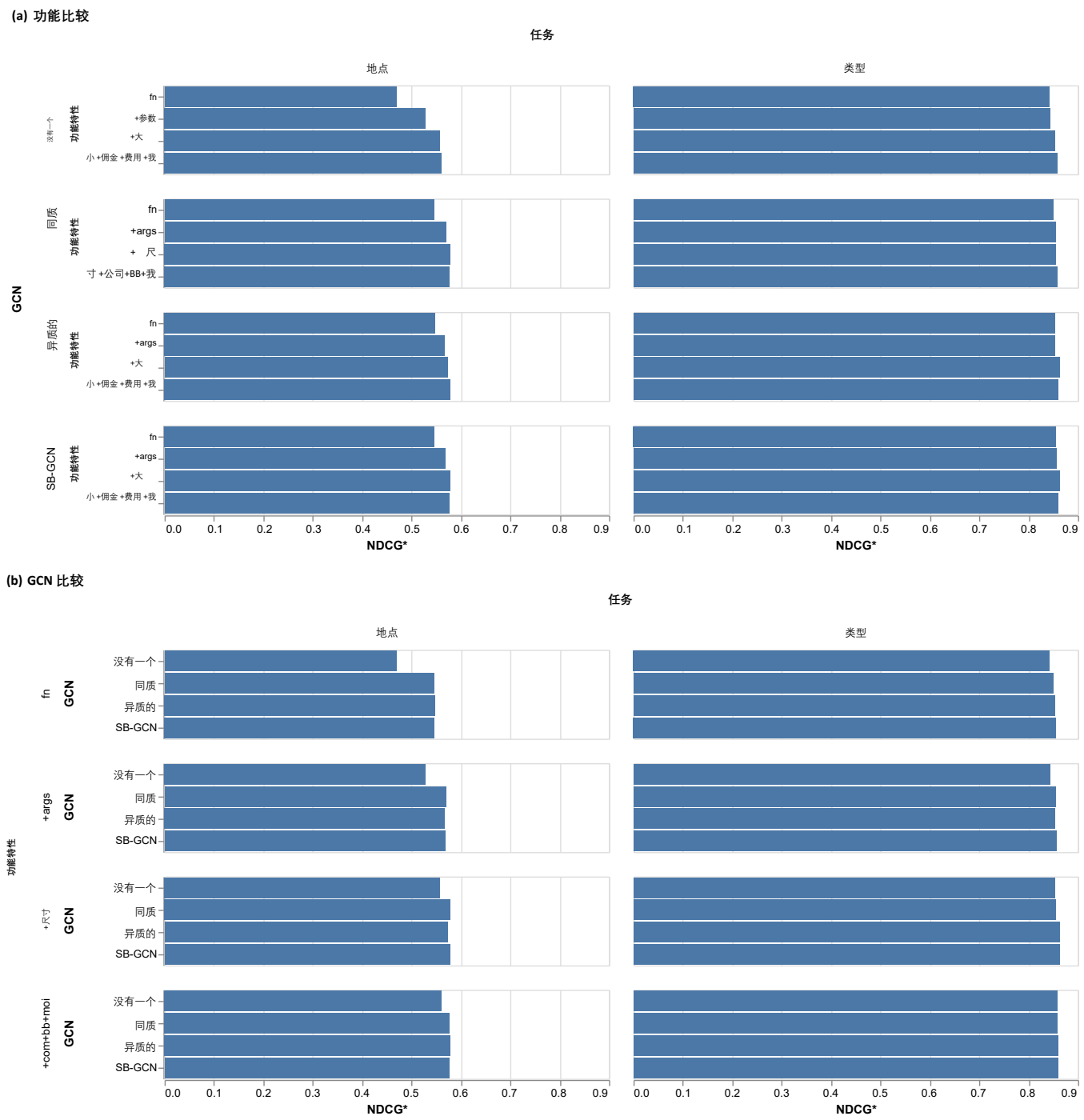


图20. 对基于70k子集训练的 GCN 类型及特征集实验进行的验证 NDCG *值。NDCG *是一种改进的 NDCG 评分，当首次匹配正确时即停止计分；对于匹配类型而言，该值与 NDCG 等价，因为仅存在一种正确的匹配类型。